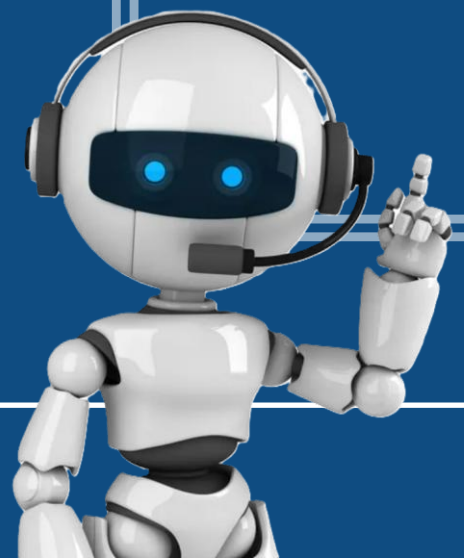
Understanding LLM Paradigm A to Z without Attention & Transformer

"All Trends up to the LLM Reasoning Era"



WOOJIN LEE woojin716@hanyang.ac.kr

1 LLM (Training) Paradigm

2 LLM Training Pipeline

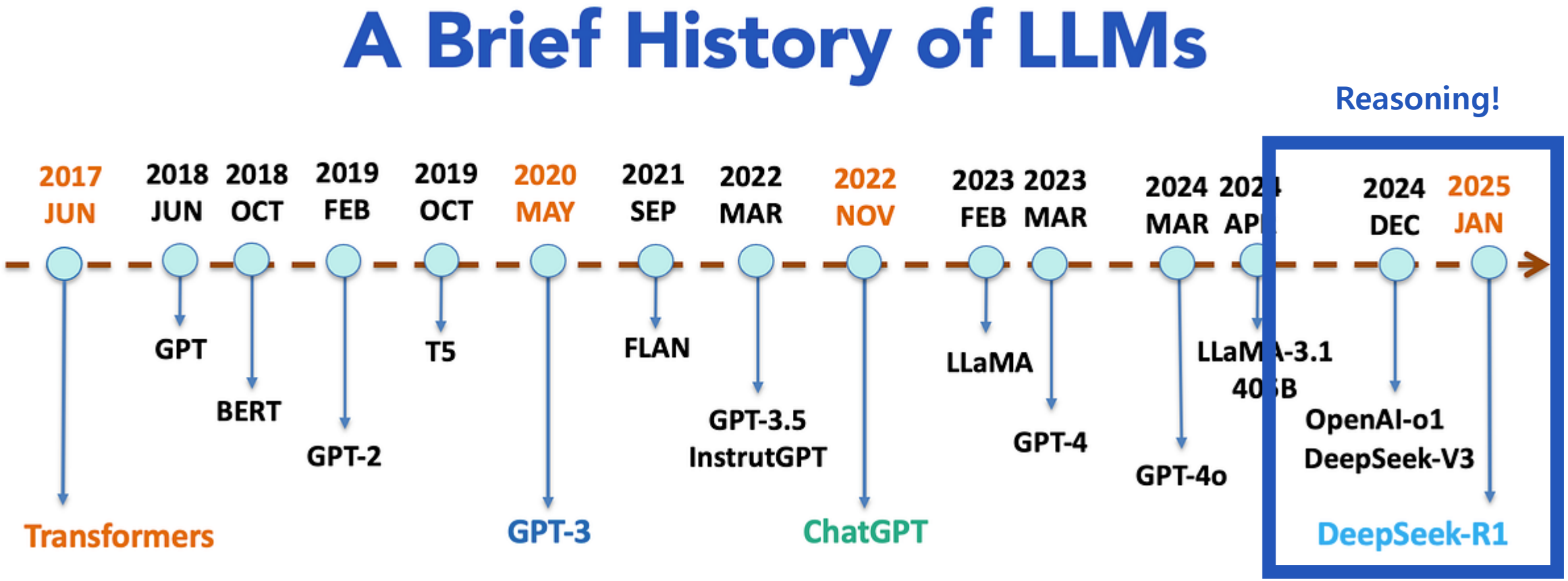
3 Understanding LLM (Training) Paradigm “Again”

4 More about LLM Reasoning

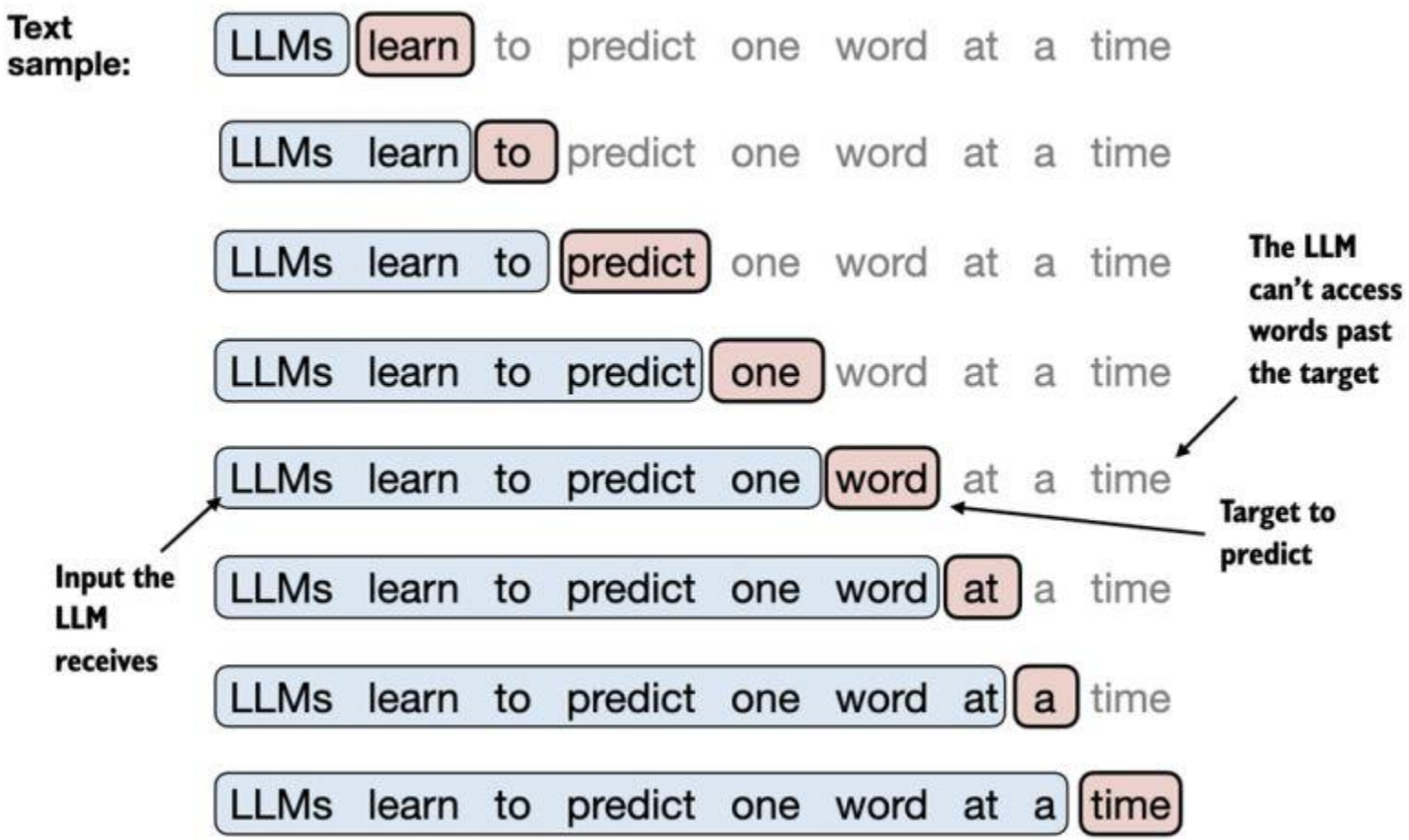
5 Now and Discussions

1. LLM (Training) Paradigm

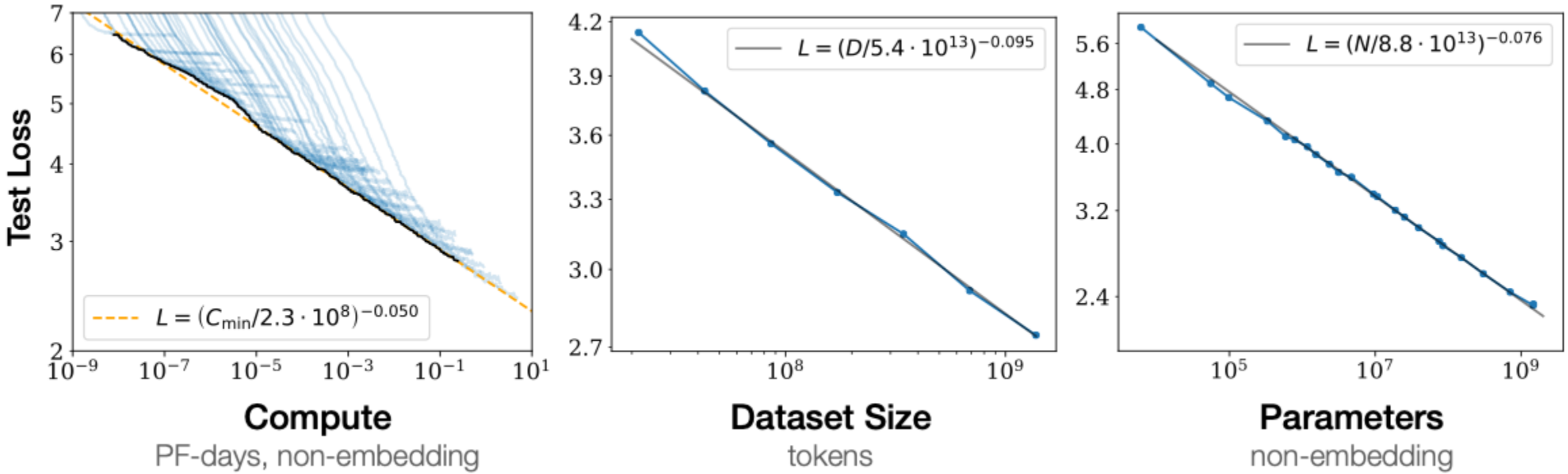
Recently… In the Era of LLMs !



Next Word Prediction (2017~)



Scaling Laws (2020)



Scaling Laws for Neural Language Models: https://arxiv.org/abs/2001.08361

In-Context Learning (Few-Shot Prompting); GPT-3 (2020)

Zero-shot

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.

```
Translate English to French:      ← task description
cheese =>                         ← prompt
```

One-shot

In addition to the task description, the model sees a single example of the task. No gradient updates are performed.

```
Translate English to French:      ← task description
sea otter => loutre de mer        ← example
cheese =>                         ← prompt
```

Few-shot

In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.

```
Translate English to French:      ← task description
sea otter => loutre de mer        ← examples
peppermint => menthe poivrée      ←
plush girafe => girafe peluche    ←
cheese =>                         ← prompt
```

Language Models are Few-Shot Learners(GPT-3): https://arxiv.org/abs/2005.14165

Chain-of-Thought Prompting (2022)

Standard Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27. ✗

Chain-of-Thought Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. 5 + 6 = 11. The answer is 11.

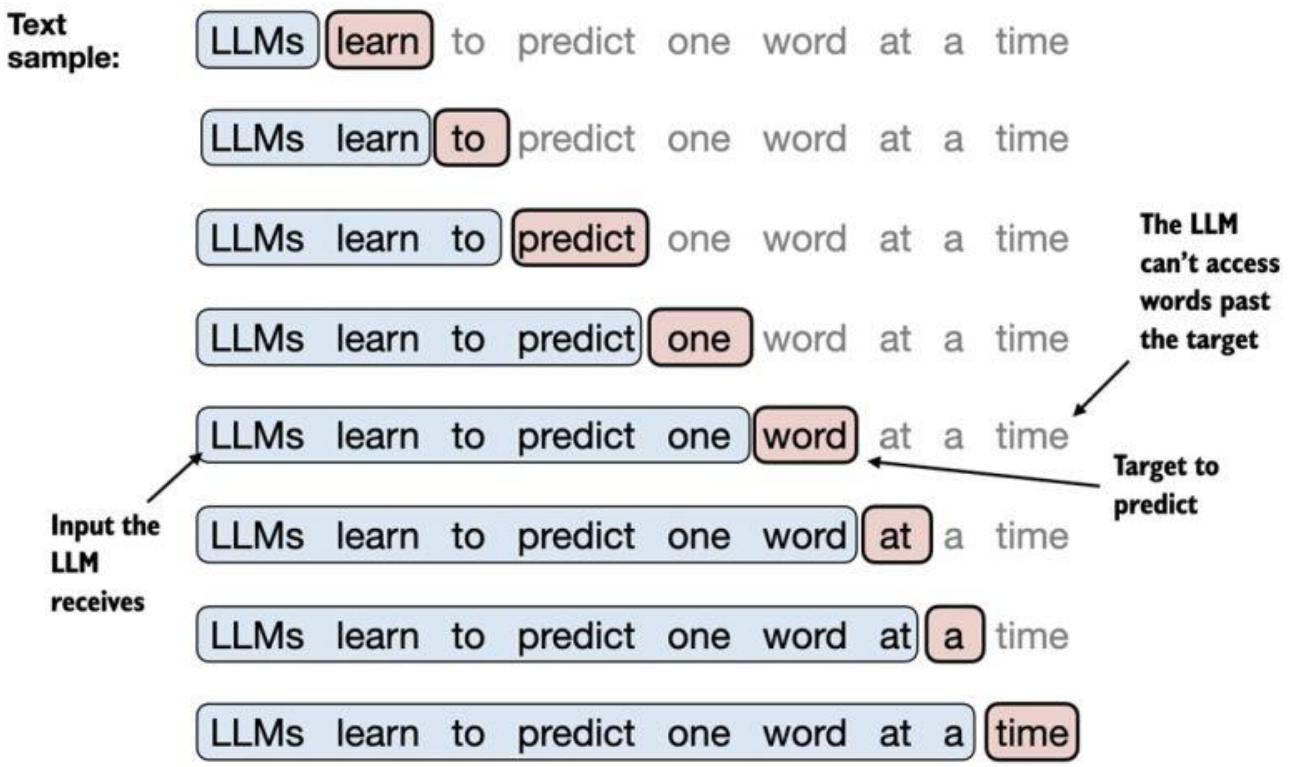
Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had 23 - 20 = 3. They bought 6 more apples, so they have 3 + 6 = 9. The answer is 9. ✓

Chain-of-Thought Prompting Elicits Reasoning in Large Language Models: https://arxiv.org/abs/2201.11903

Again, Next Word Prediction (2017~) + Prompt Engineering



Few-shot

In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.

```
Translate English to French:
sea otter => loutre de
peppermint => menthe p
plush girafe => girafe
cheese =>
```

task description

Chain-of-Thought Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

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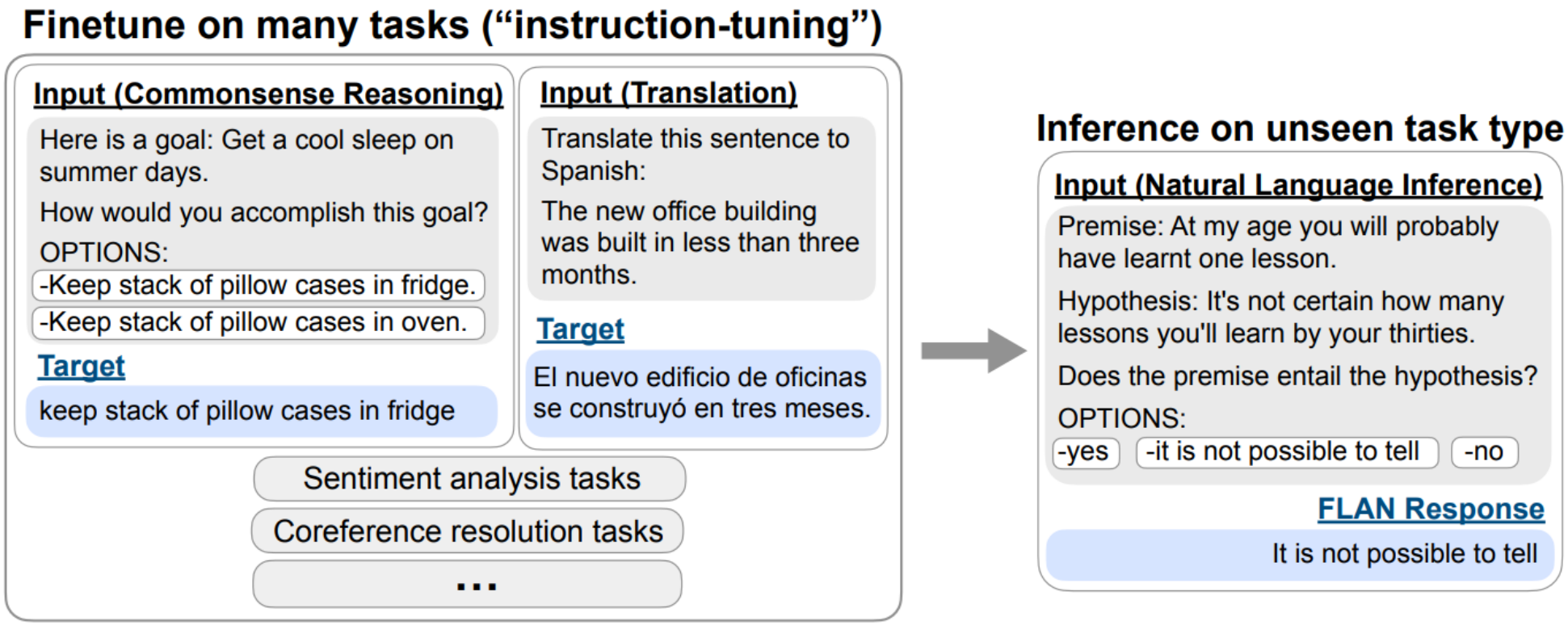
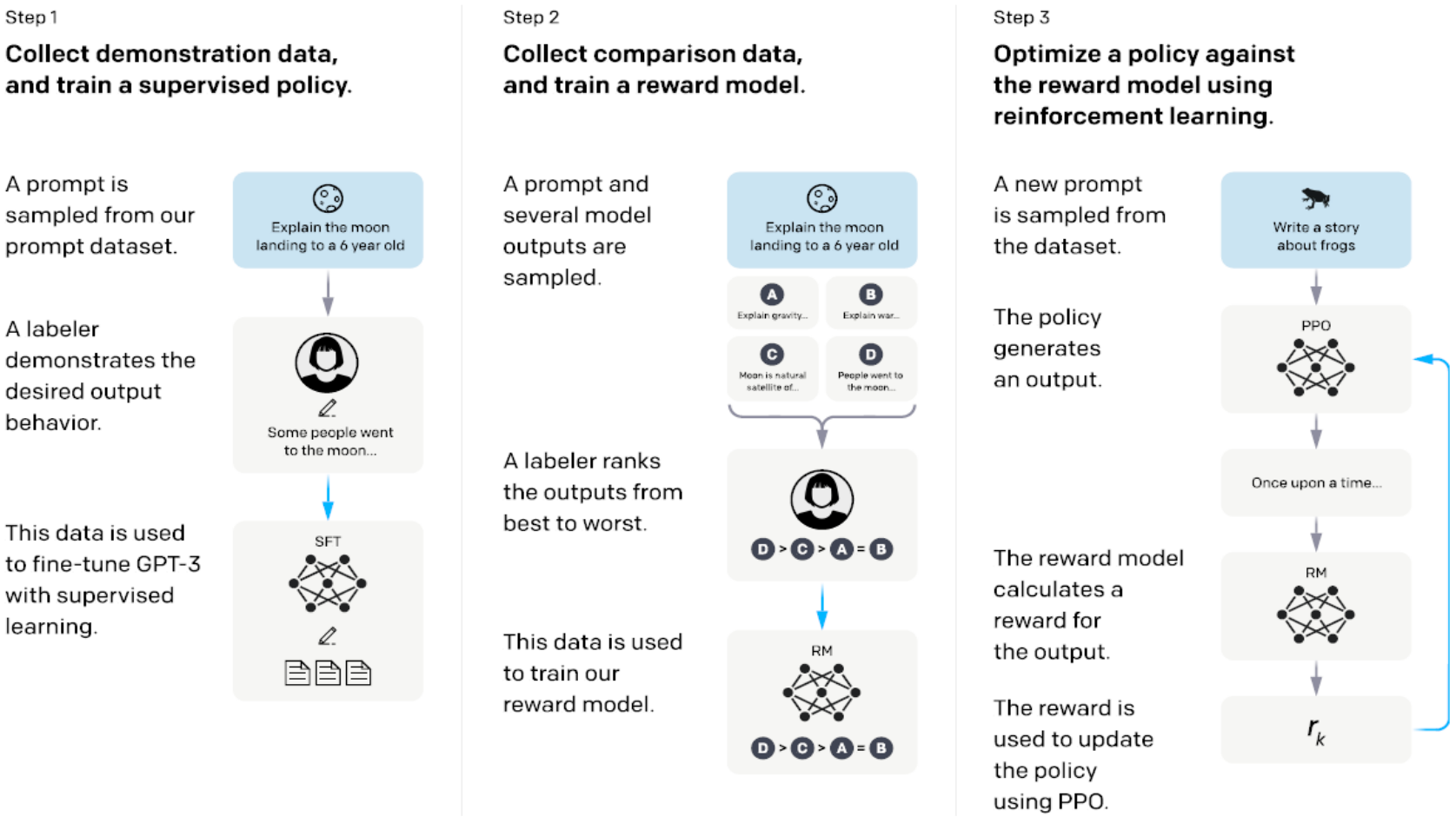
Instruction-Tuning (FLAN-2021)



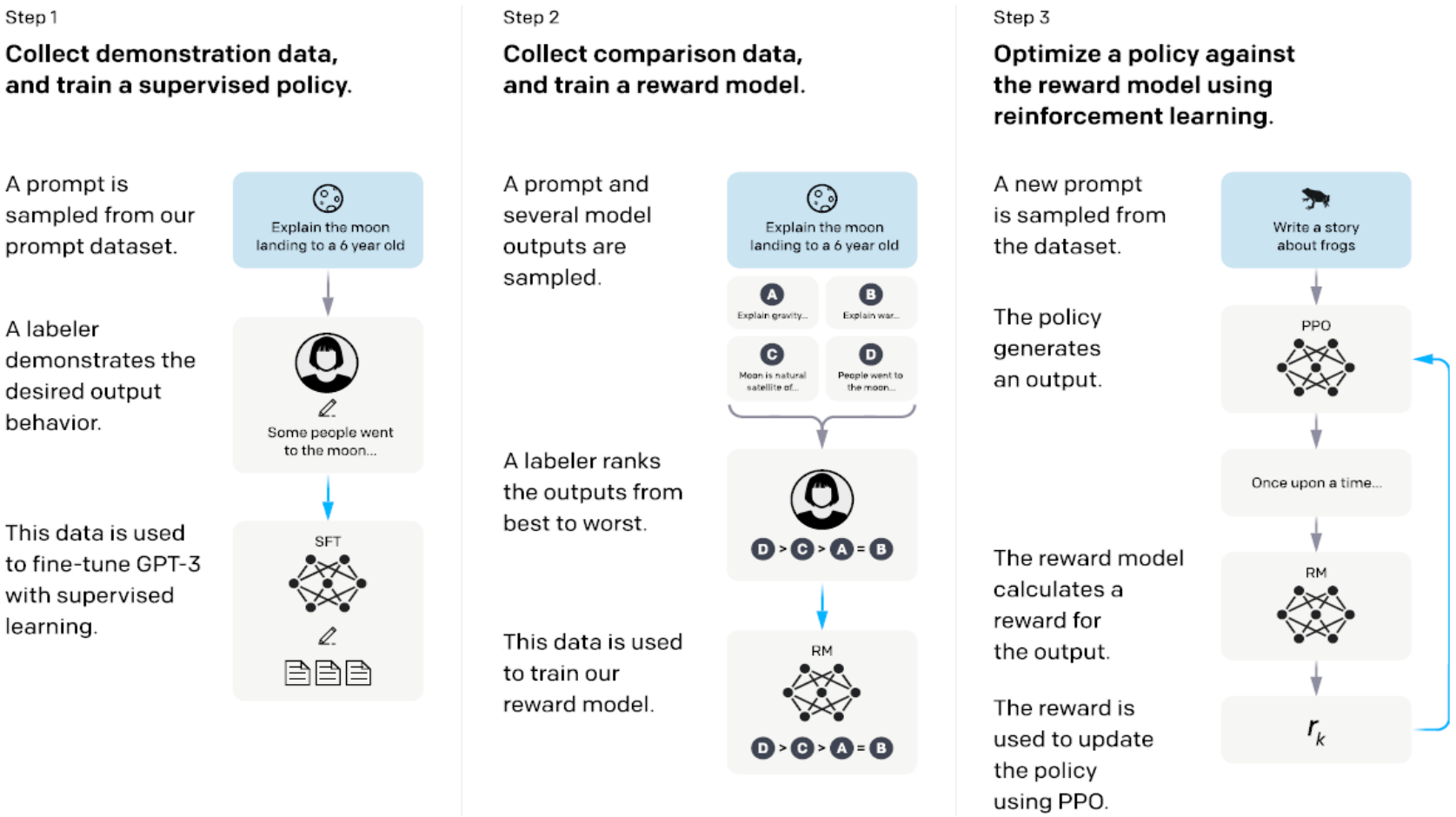
Finetuned Language Models Are Zero-Shot Learners(FLAN): https://arxiv.org/abs/2109.01652

Instruction-Tuning (InstructGPT-2022)



Training language models to follow instructions with human feedback(InstructGPT): https://arxiv.org/abs/2203.02155

Reinforcement Learning from Human Feedback; RLHF (PPO) (2022)



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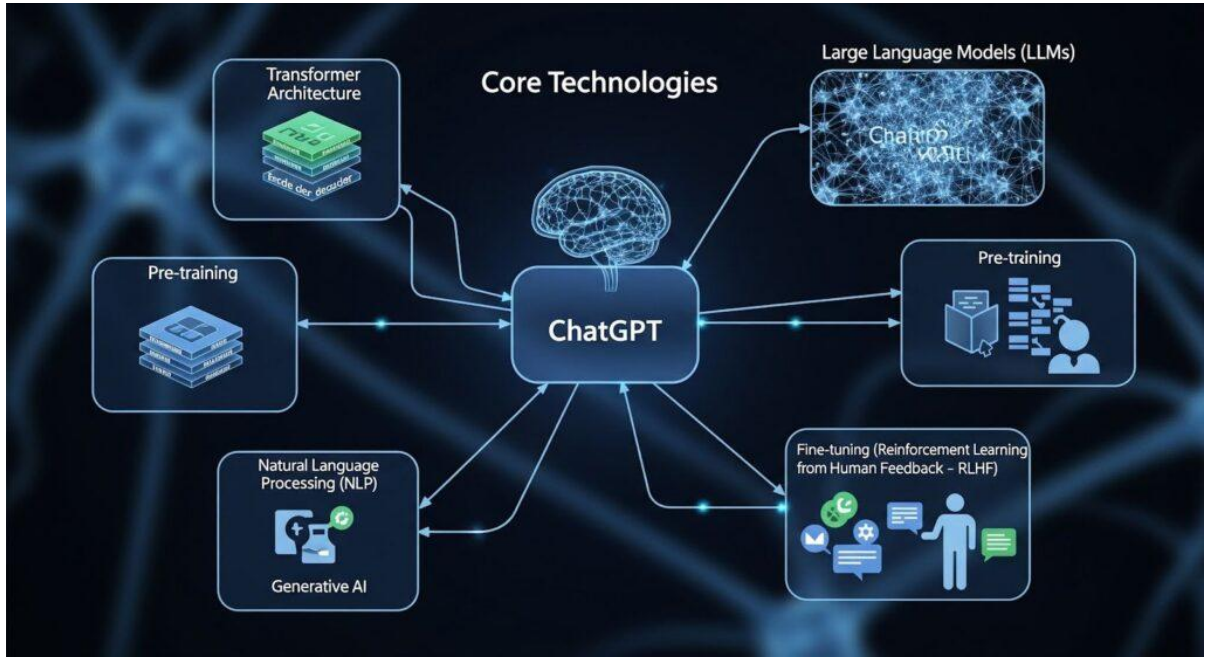
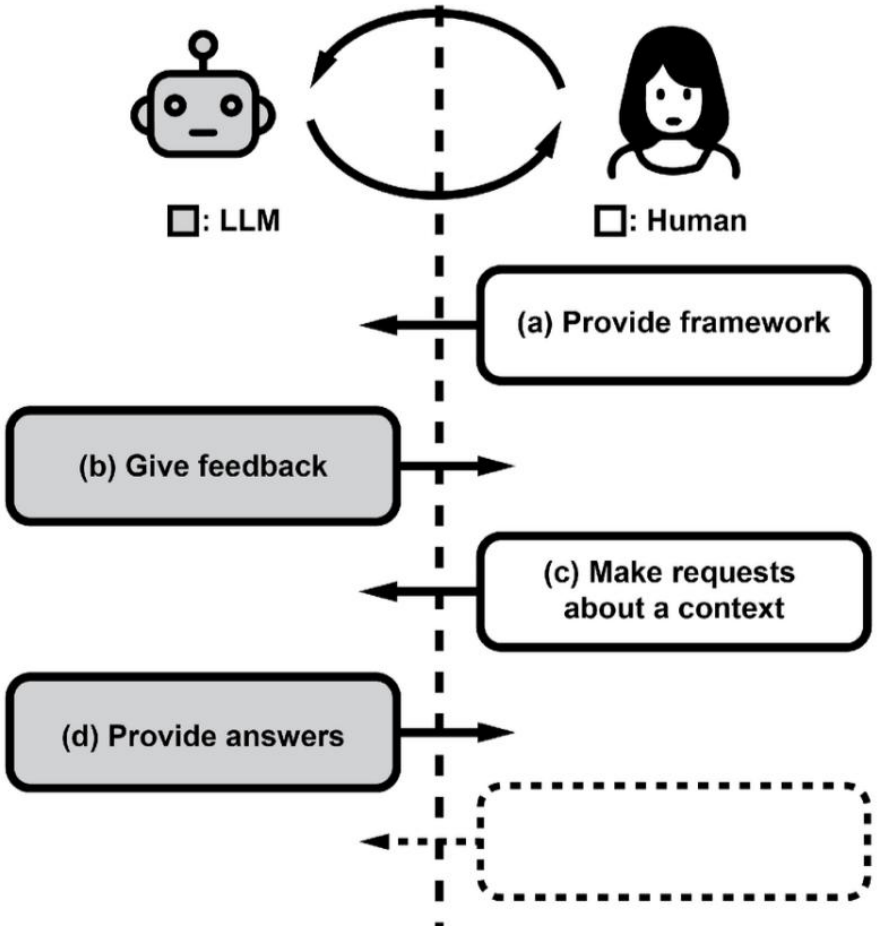
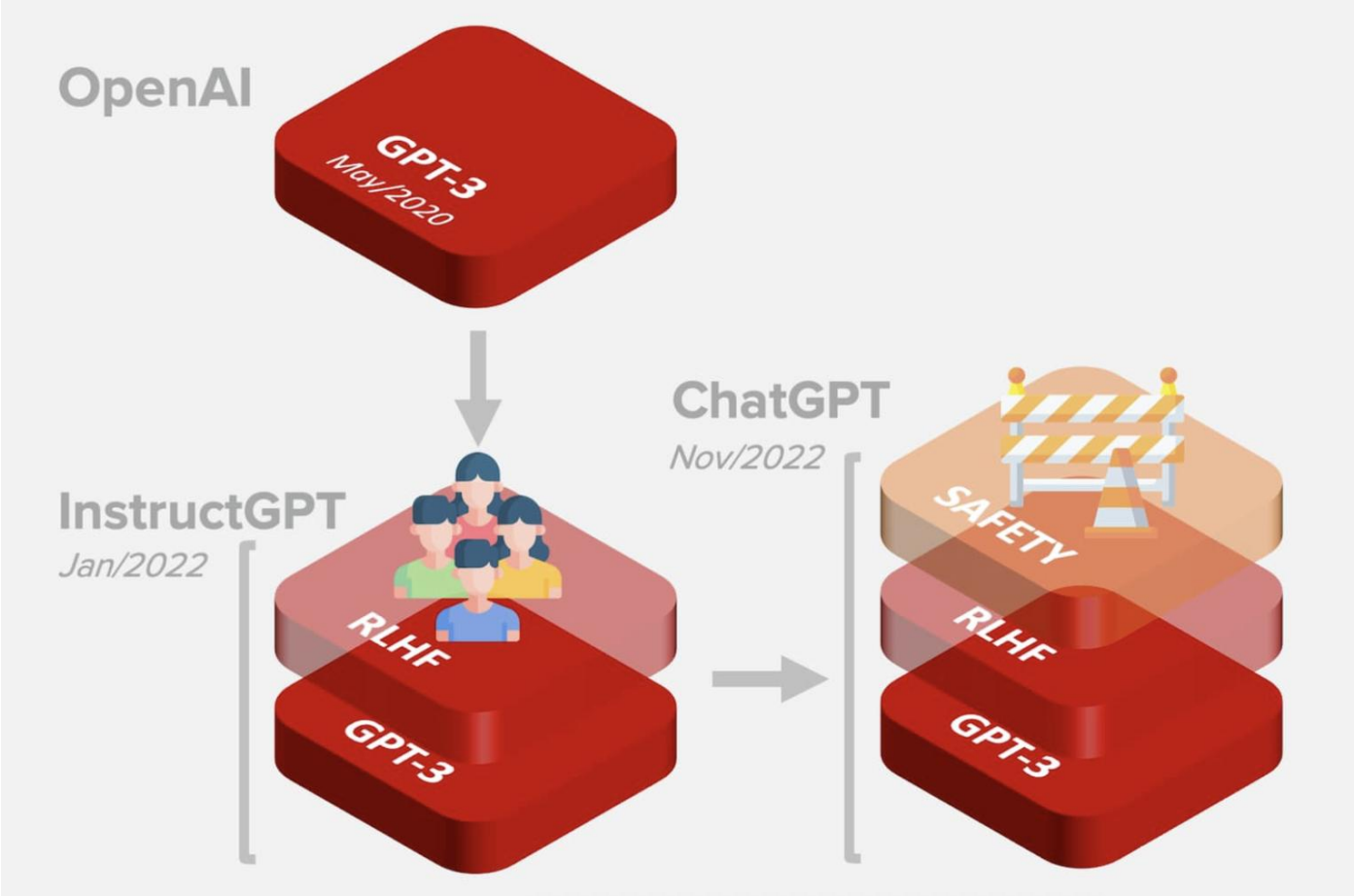
From Next Word Prediction (2017~) + Prompt Engineering to Instruction-Following (2021~) & Preference Alignment (2022~)

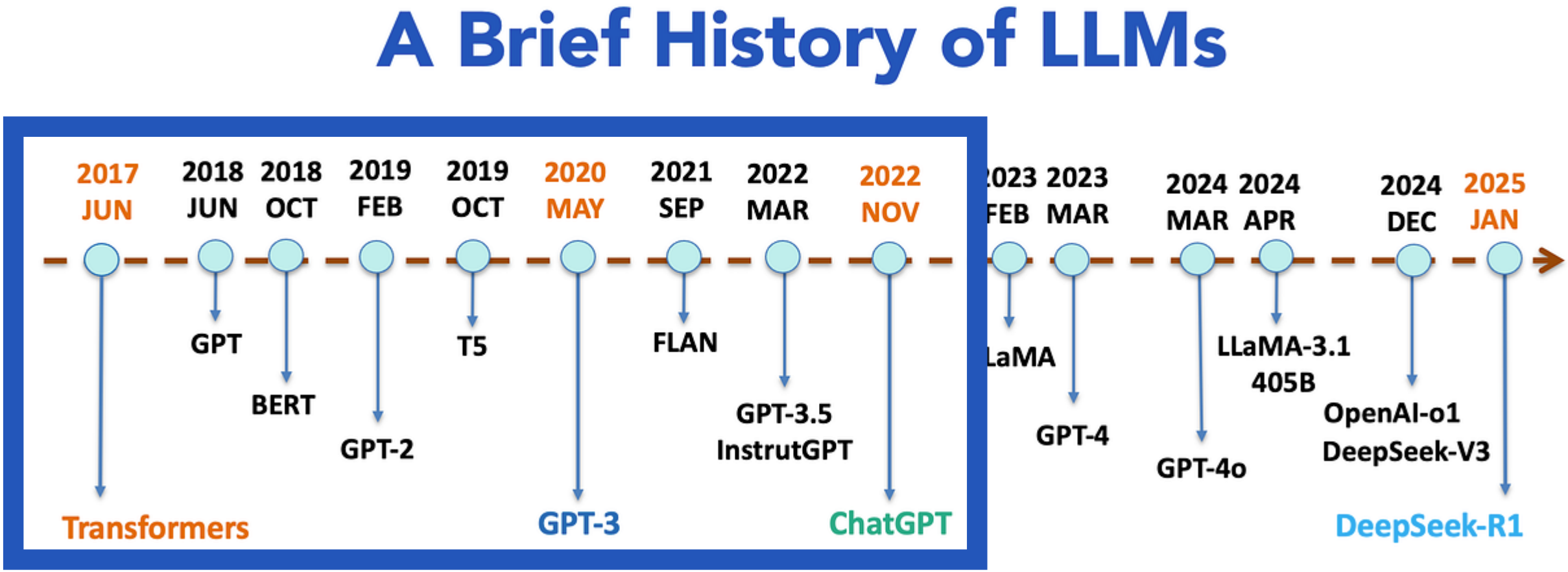




ChatGPT (2022)



Again, In the Era of LLMs



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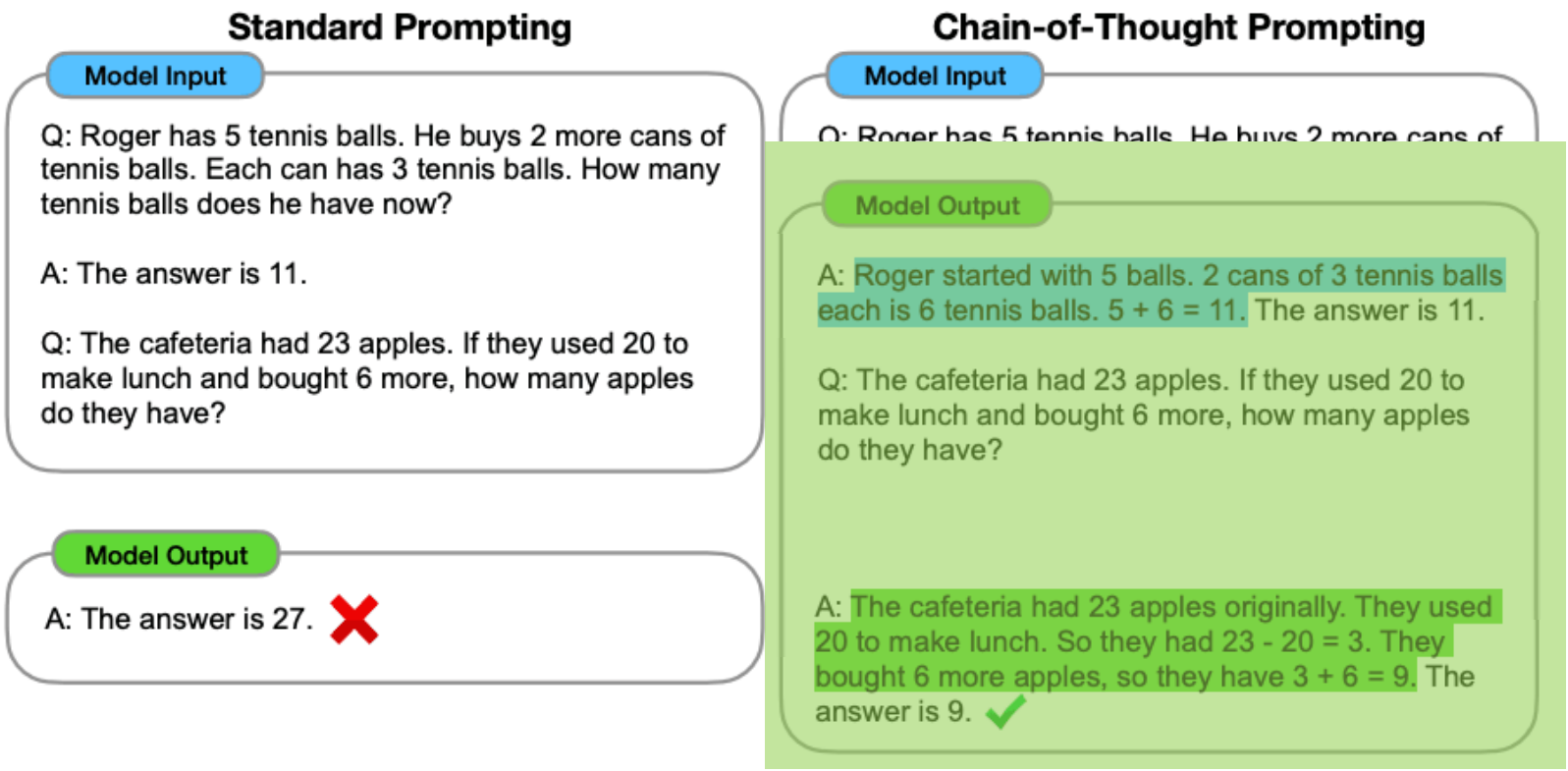
Now, Chain-of-Thought ~~Prompting~~ Generation (2022)



Large Language Models are Zero-Shot Reasoners: https://arxiv.org/abs/2205.11916

Now, Chain-of-Thought ~~Prompting~~ Generation (2022)

(a) Few-shot

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?
A: The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A:

(Output) The answer is 8. ✗

(b) Few-shot-CoT

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?
A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. 5 + 6 = 11. The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A:

(Output) The juggler can juggle 16 balls. Half of the balls are golf balls. So there are 16 / 2 = 8 golf balls. Half of the golf balls are blue. So there are 8 / 2 = 4 blue golf balls. The answer is 4. ✓

(c) Zero-shot

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A: The answer (arabic numerals) is

(Output) 8 ✗

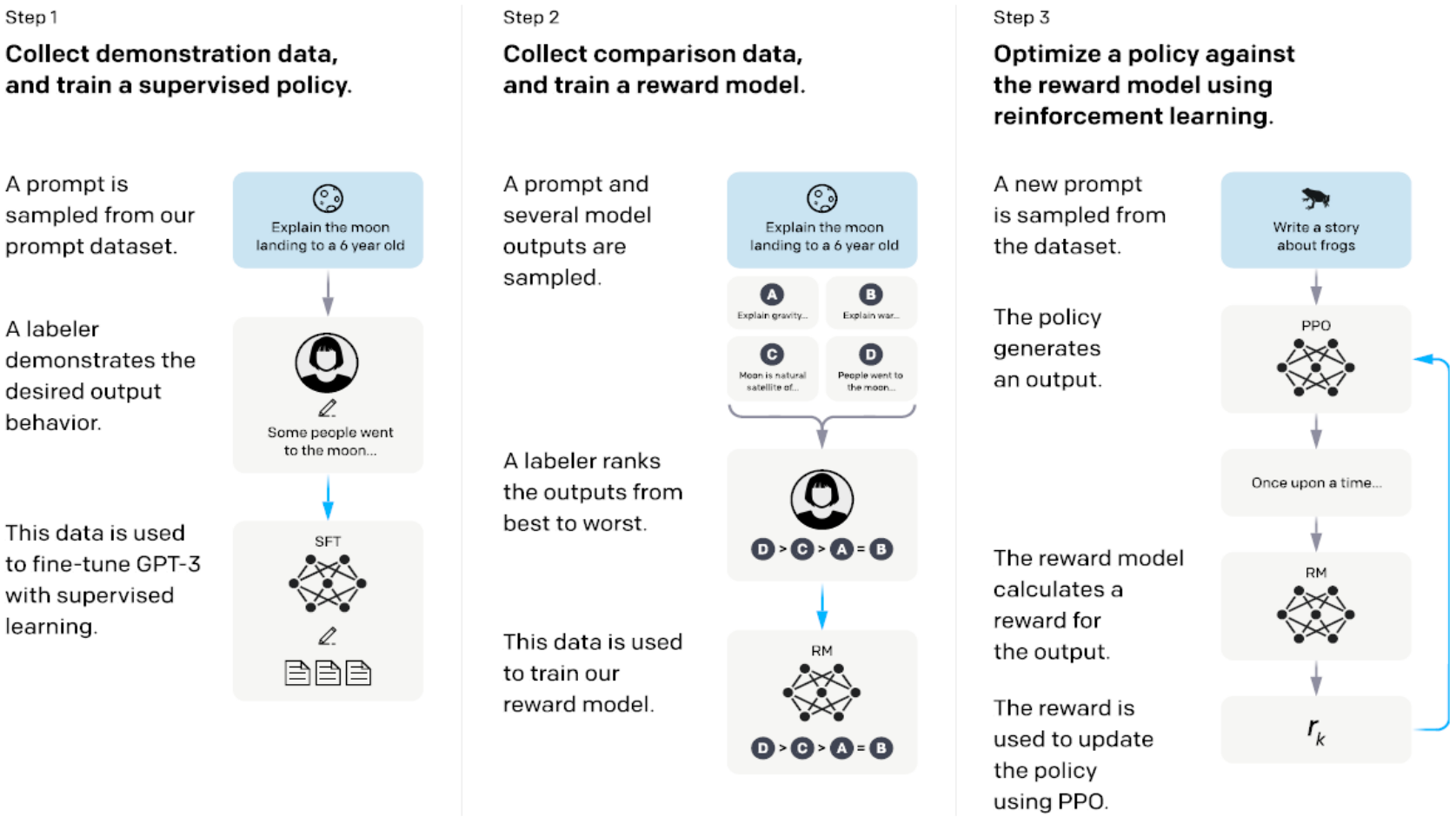
(d) Zero-shot-CoT (Ours)

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A: ***Let's think step by step.***

(Output) There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls. ✓

Large Language Models are Zero-Shot Reasoners: https://arxiv.org/abs/2205.11916

Reinforcement Learning from Human Feedback; RLHF (PPO) (2022)



Training language models to follow instructions with human feedback(InstructGPT): https://arxiv.org/abs/2203.02155

Direct Preference Optimization; DPO (2023)

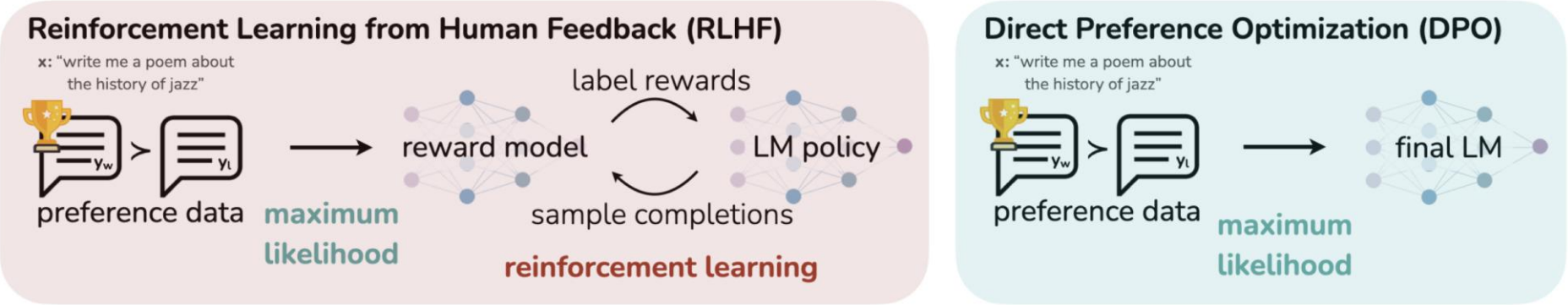
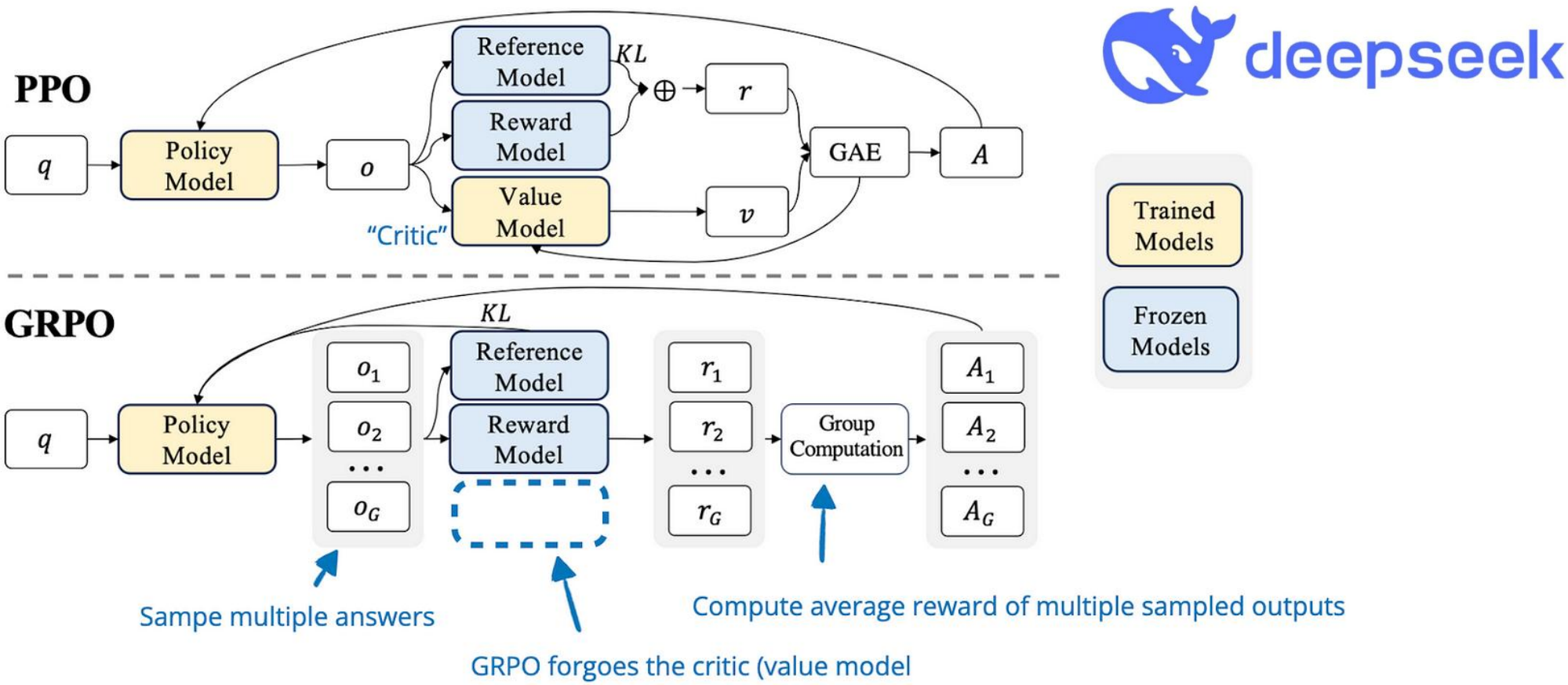


Figure 1: **DPO optimizes for human preferences while avoiding reinforcement learning.** Existing methods for fine-tuning language models with human feedback first fit a reward model to a dataset of prompts and human preferences over pairs of responses, and then use RL to find a policy that maximizes the learned reward. In contrast, DPO directly optimizes for the policy best satisfying the preferences with a simple classification objective, without an explicit reward function or RL.

Direct Preference Optimization: Your Language Model is Secretly a Reward Model: https://arxiv.org/abs/2305.18290

Group Relative Policy Optimization; GRPO (DeepSeekMath-2024)



DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models: https://arxiv.org/abs/2402.03300

Reinforcement Learning with Verifiable Rewards; RLVR (Tulu3-2024)

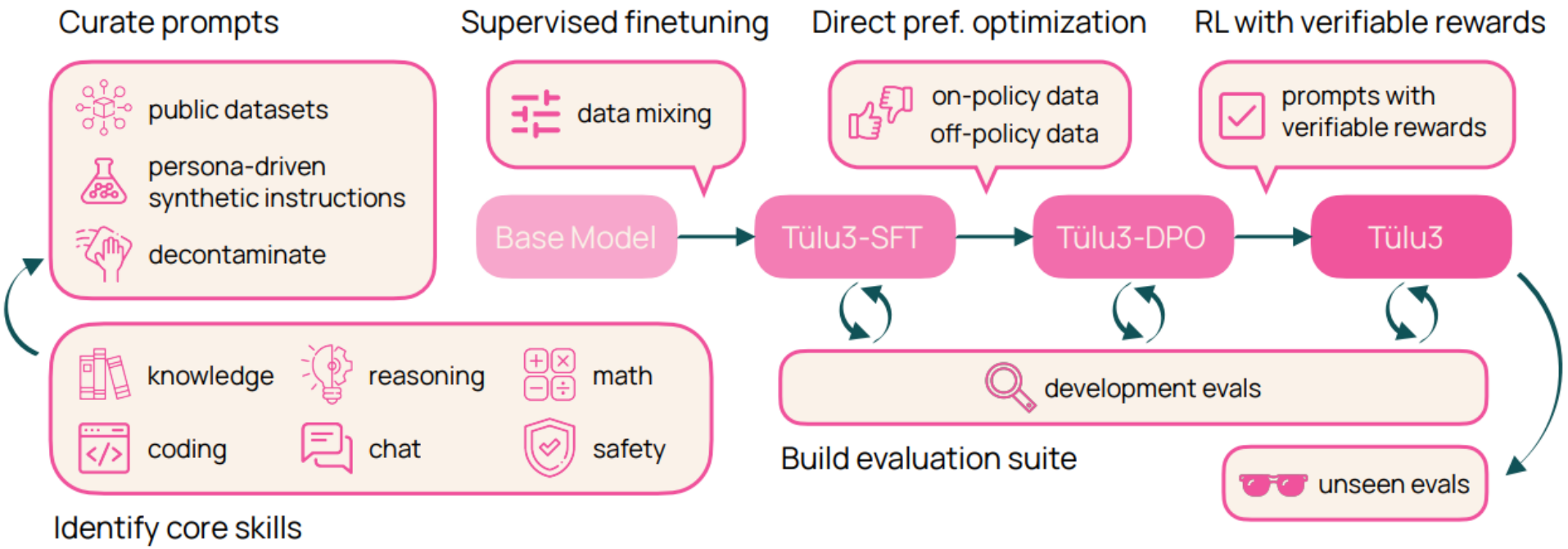
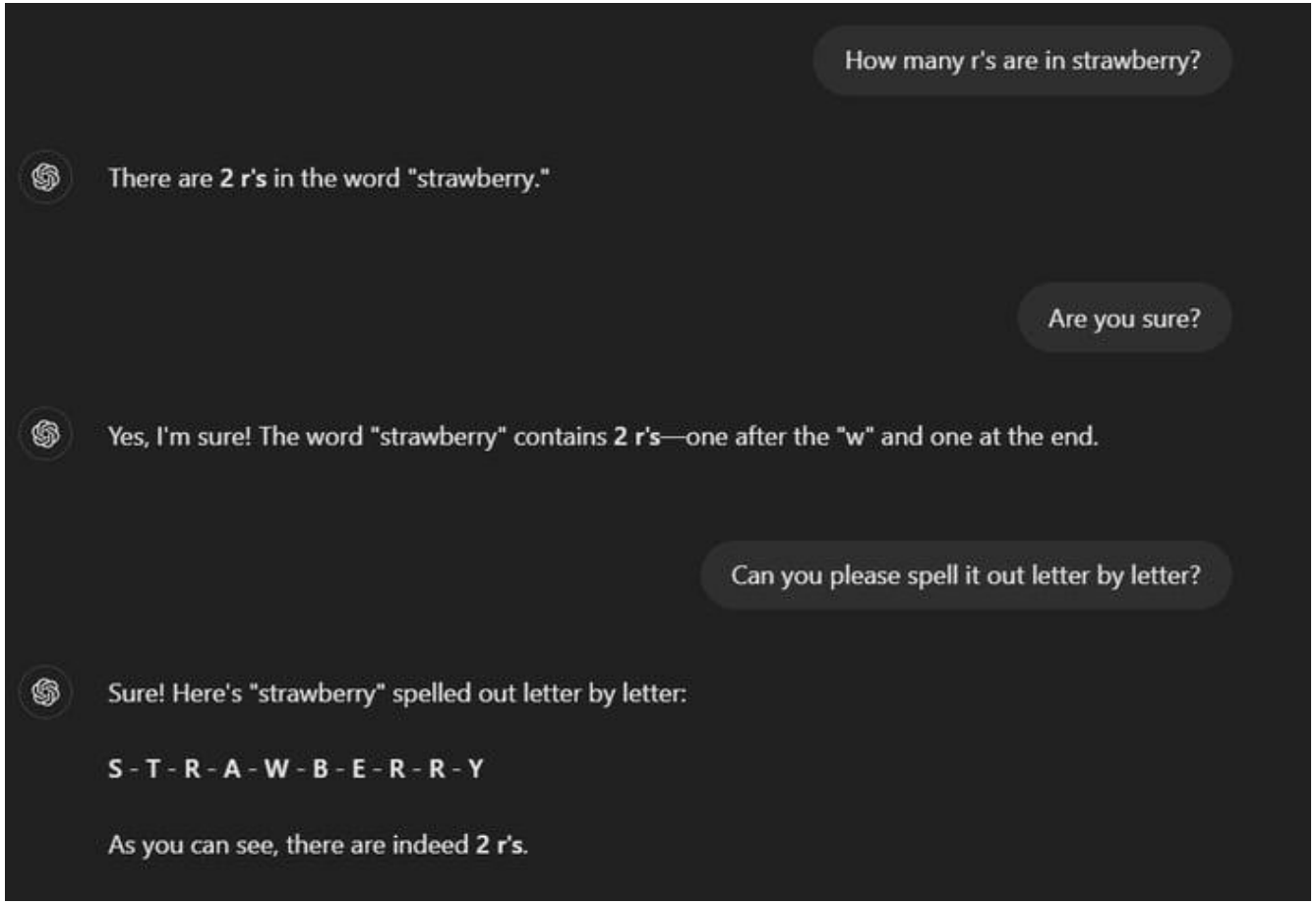


Figure 1 An overview of the TÜLU 3 recipe. This includes: data curation targeting general and target capabilities, training strategies and a standardized evaluation suite for development and final evaluation stage.

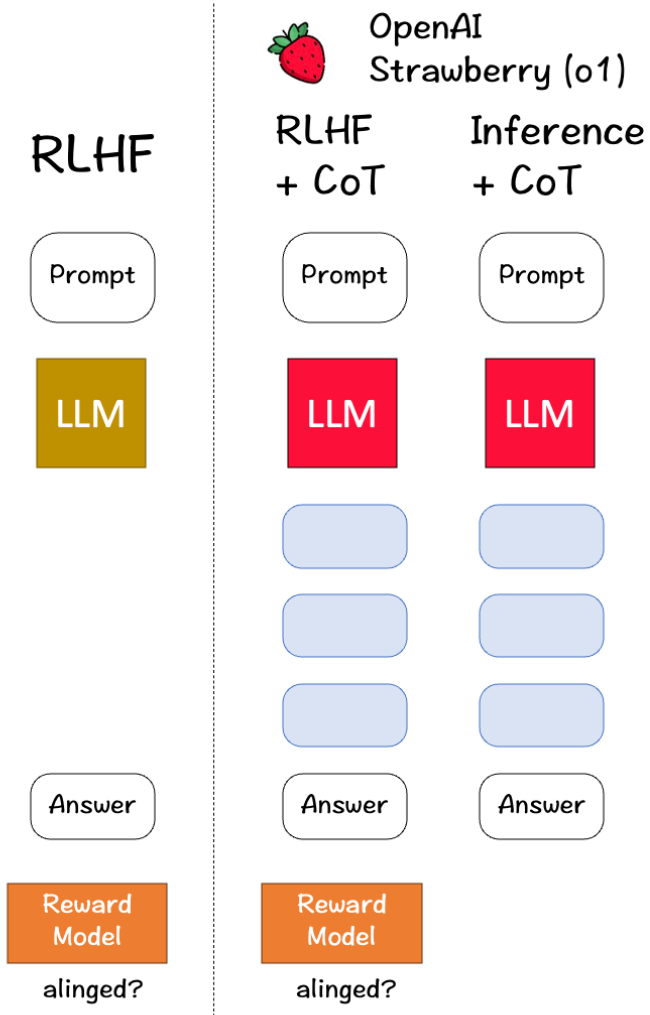
Tulu 3: Pushing Frontiers in Open Language Model Post-Training: https://arxiv.org/abs/2411.15124

Inference/Test-Time Scaling (OpenAI-o1, a.k.a. strawberry-2024)



OpenAI o1 System Card: https://arxiv.org/abs/2412.16720

Inference/Test-Time Scaling (OpenAI-o1, a.k.a. strawberry-2024)



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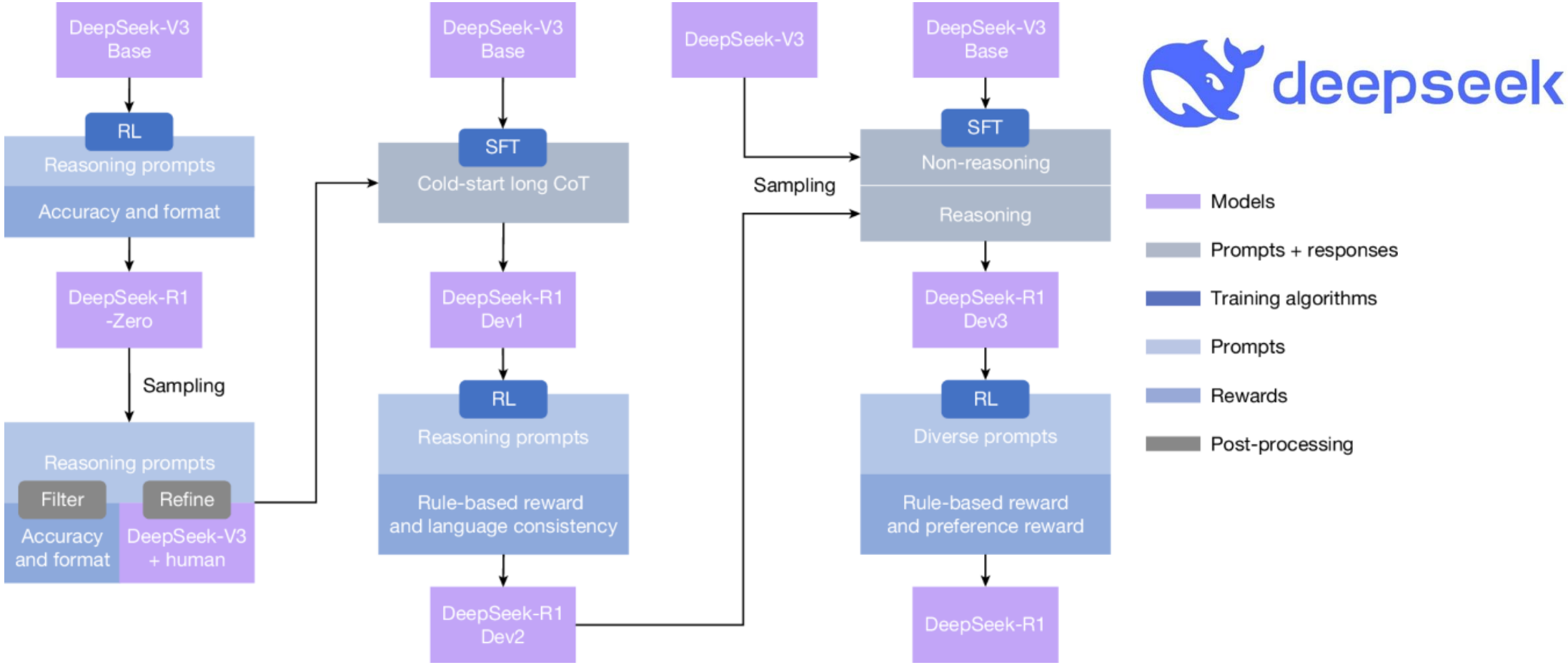
Multi-Stage RL, Catching up to BigTech; OpenAI-o1 (DeepSeek-R1, 2025)



DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning: https://arxiv.org/abs/2501.12948

Multi-Stage RL, Catching up to BigTech; OpenAI-o1 (DeepSeek-R1, 2025)

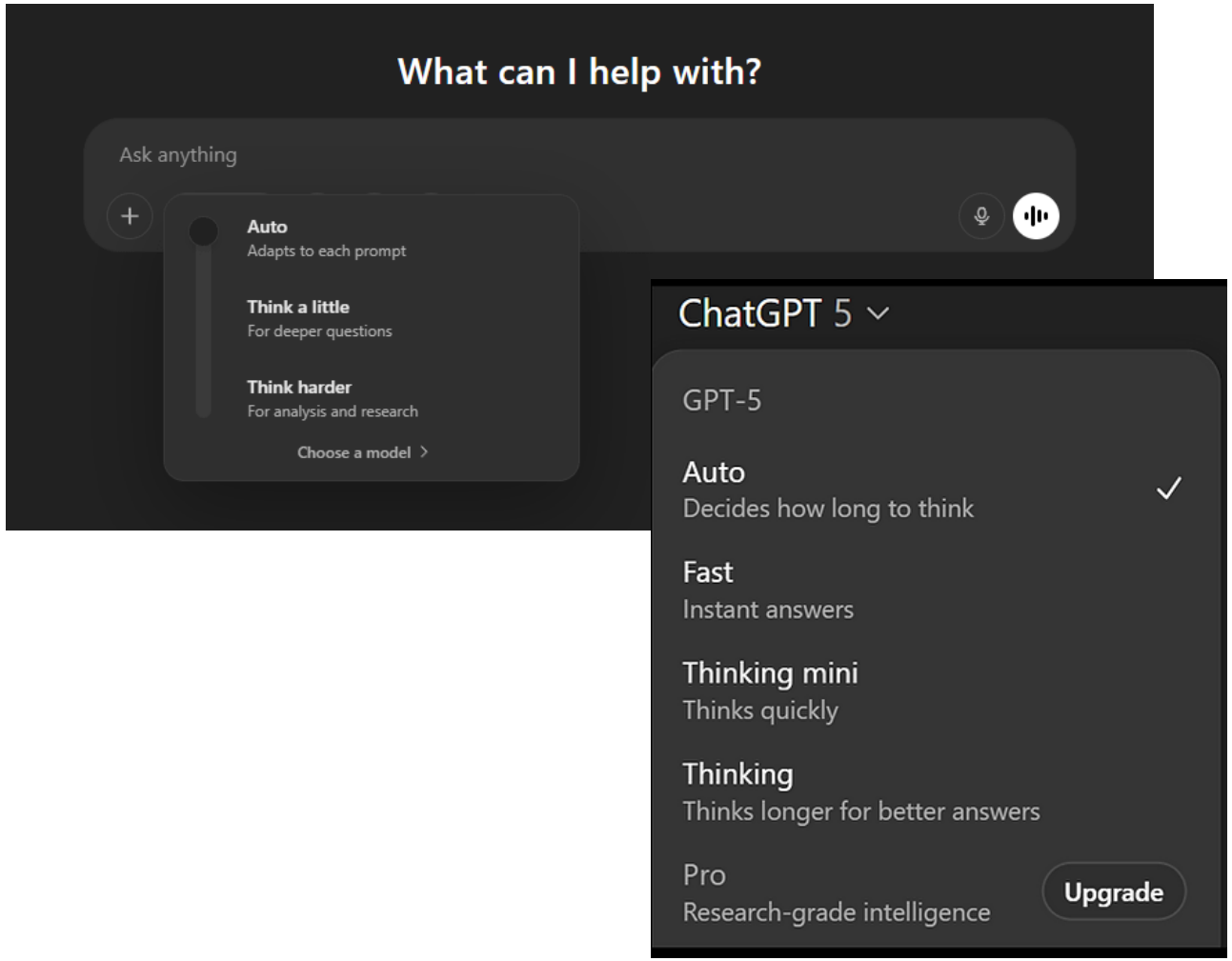
Question: If $a > 1$, then the sum of the real solutions of $\sqrt{a - \sqrt{a + x}} = x$ is equal to

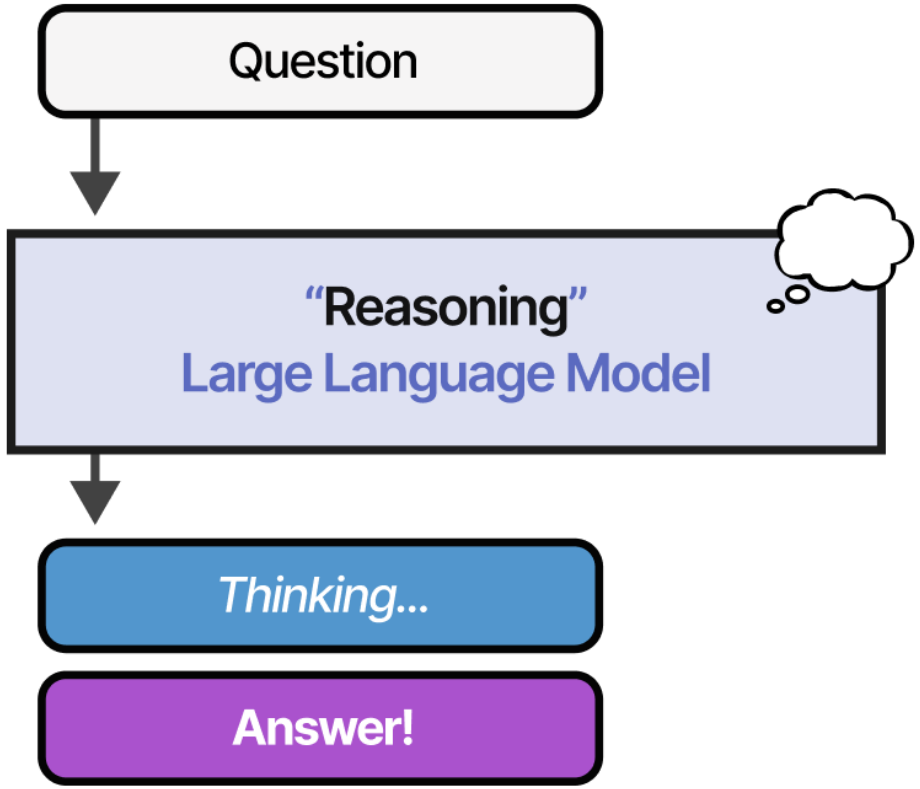
Response: <think>
To solve the equation $\sqrt{a - \sqrt{a + x}} = x$, let's start by squaring both $\cdots$
$\left(\sqrt{a - \sqrt{a + x}}\right)^2 = x^2 \implies a - \sqrt{a + x} = x^2.$
Rearrange to isolate the inner square root term:
$(a - x^2)^2 = a + x \implies a^2 - 2ax^2 + (x^2)^2 = a + x \implies x^4 - 2ax^2 - x + (a^2 - a) = 0$
$\ldots$
Wait, wait. Wait. That's an aha moment I can flag here.
Let's reevaluate this step-by-step to identify if the correct sum can be $\cdots$
We started with the equation:
$\sqrt{a - \sqrt{a + x}} = x$
First, let's square both sides:
$a - \sqrt{a + x} = x^2 \implies \sqrt{a + x} = a - x^2$
Next, I could square both sides again, treating the equation: $\cdots$
$\ldots$

Table 3 | An interesting "aha moment" of an intermediate version of DeepSeek-R1-Zero. The model learns to rethink using an anthropomorphic tone. This is also an aha moment for us, allowing us to witness the power and beauty of reinforcement learning.

DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning: https://arxiv.org/abs/2501.12948

From Instruction-Following (2021~) & Preference Alignment (2022~) to Reasoning Capability (2024~)





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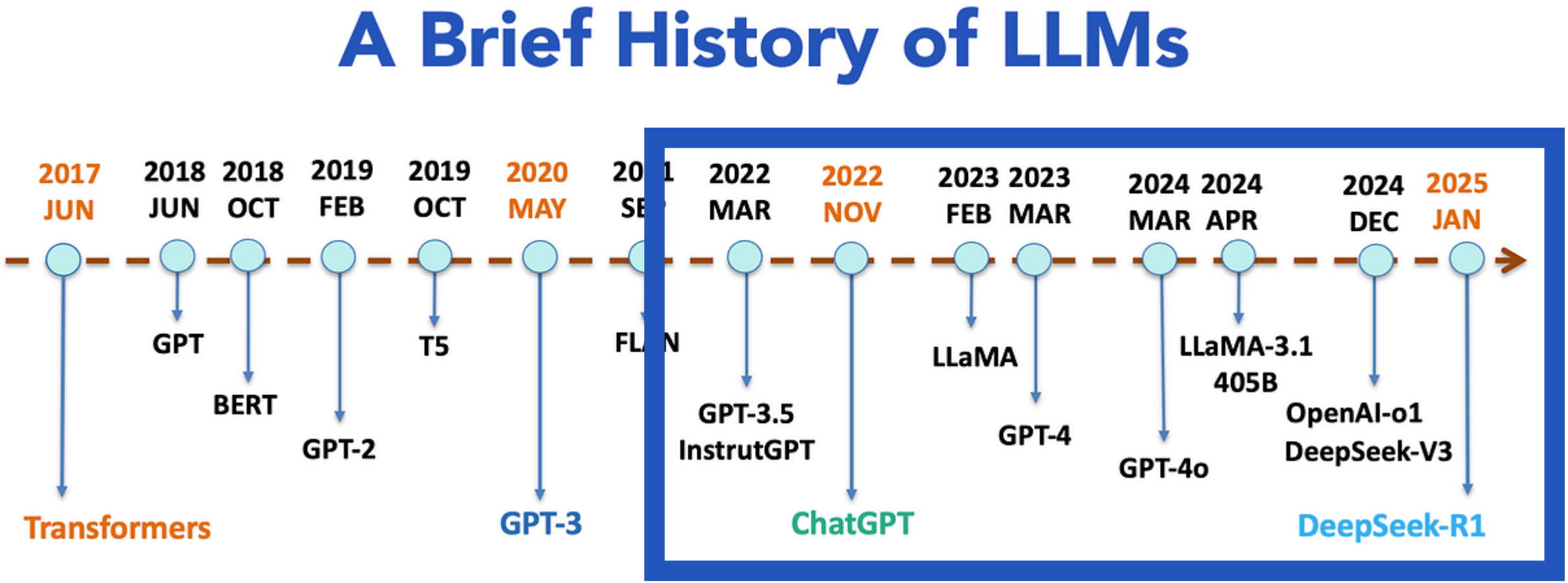


2. LLM Training Pipeline

Again, LLM (Training) Paradigm

LLM Training Objective	Keywords
Next Word Prediction (2017~)	Scaling Laws In-Context Learning (Few-Shot) Chain-of-Thought Prompting
+	+
Instruction Following (2021~) Preference Alignment (2022~)	Instruction-Tuning Supervised Fine-Tuning RLHF, DPO
+	+
Reasoning Capability (2024~)	GRPO, RLVR Inference/Test-Time Scaling Reasoning

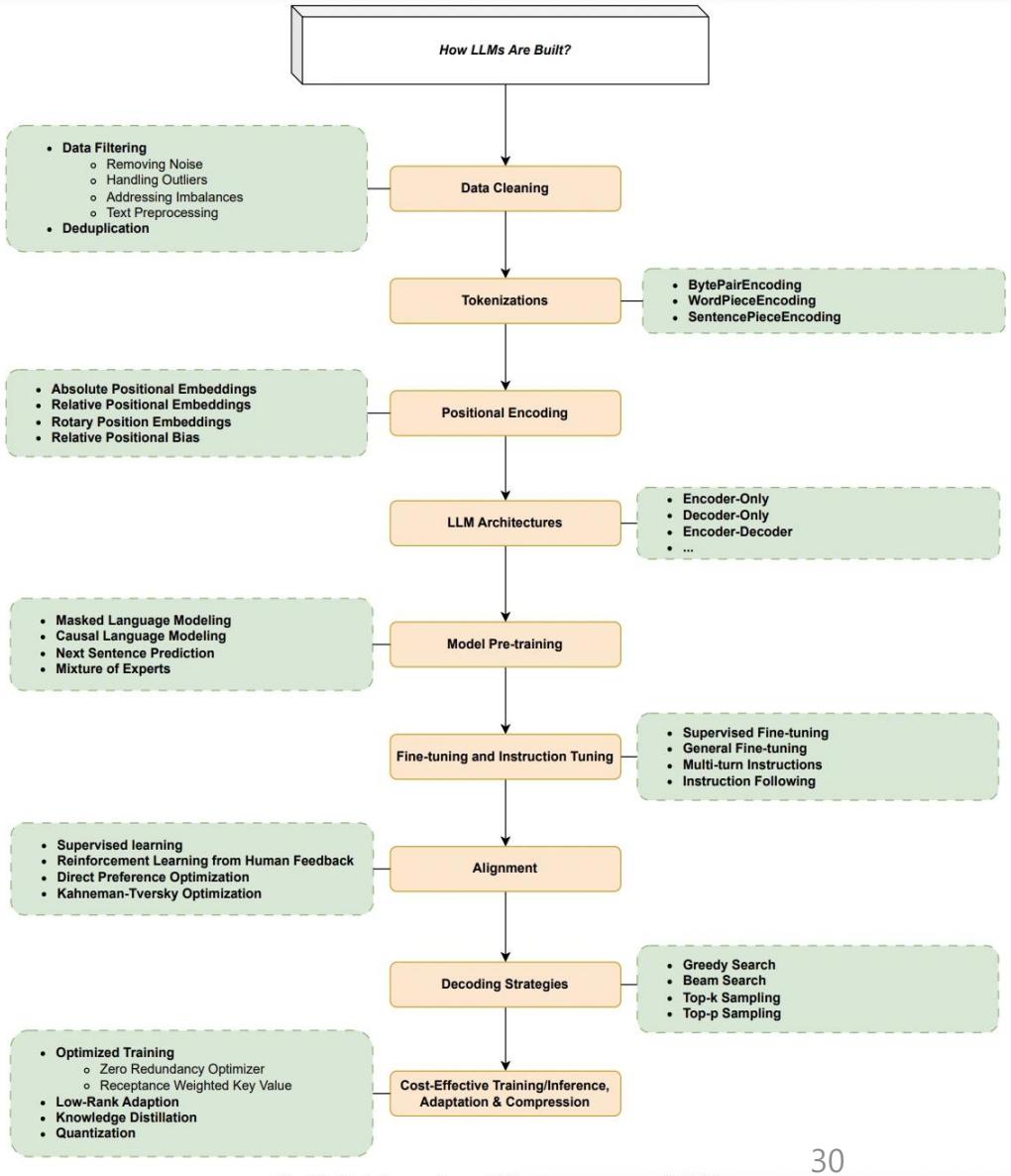


Fig. 25: This figure shows different components of LLMs.

LLM Training, “From Scratch”

독자 AI 파운데이션 모델 프로젝트 5개 정예팀

1 목표 2 컨소시엄 ※컨소시엄 주관사 기준 가나다순, 자료: 과학기술정보통신부

NAVER Cloud

1 AI 접근성 확대 위한 옴니 파운데이션 모델 원천기술 연구·개발

2 네이버·트웰브랩스·서울대·카이스트(KAIST)·포항공대·고려대·한양대

upstage

1 글로벌 프런티어 파운데이션 모델 개발

2 노타·래블업·플리토·뷰노·마키나락스·로앤컴퍼니·오케스트로·데이원컴퍼니·올거나이즈코리아·금융결제원·서강대·카이스트

SK telecom

1 언어·멀티모달·행동 융합 차세대 트랜스포머 기반 초거대 모델 개발

2 크래프톤·포티투닷·리벨리온·라이너·셀렉트스타·서울대·카이스트

NC AI

1 산업 AI 전환 위한 멀티모달 생성용 파운데이션 모델 개발

2 고려대·서울대·연세대·카이스트·한국전자통신연구원·에이아이웍스·포스코DX·롯데이노베이트·HL로보틱스·인터엑스·미디어젠·MBC·NHN

LG AI Research

1 글로벌 최고 수준 프런티어 AI 모델 ‘K-엑사원’ 개발

2 LG유플러스·LG CNS·슈퍼브AI·퓨리오사AI·프렌들리AI·이스트소프트·이스트에이드·한글과컴퓨터·뤼튼테크놀로지스

프로젝트 개요 | 6개월 단위 성과 측정, 2027년 2개 팀으로 압축 | 글로벌 최상위 AI 모델 대비 95% 이상 성능 확보 | 그래픽처리장치(GPU)·데이터·인건비 지원 | ‘K-AI 모델·기업’ 명칭 부여 | 각 팀 개발 AI 모델의 생태계 확장

‘독자 인공지능(AI) 파운데이션 모델’ 5개 컨소시엄

당락	컨소시엄	참여 기업·기관·대학·연구팀
2단계 진출	LG AI연구원	LG유플러스, LG CNS, 퓨리오사AI, 한글과컴퓨터, 뤼튼테크놀로지스 등
	SK텔레콤	크래프톤, 포티투닷, 리벨리온, 라이너, 셀렉트스타 등
	업스테이지	노타, 래블업, 플리토, 뷰노, 마키나락스 등
탈락	네이버 클라우드	트웰브랩스, 서울대 산학협력단 등
	NC AI	포스코DX, 고려대, 서울대, 연세대, KAIST 등

독자 AI 파운데이션 모델 프로젝트 1차 결과

K-AI

2차 단계 진출

LG AI연구원	벤치마크, 전문가, 사용자 평가 전 부문에서 최고점 기록 LG AI연구원
SKT	NIA 벤치마크 및 개별 벤치마크 부문에서 최고 수준의 성적 거둠 SK텔레콤
업스테이지	upstage

진출 실패

네이버 클라우드	모델 설계 및 사전학습 과정 ‘독자성’ 기준 미충족에 따른 부적격 판정 NAVER Cloud
NC AI	종합 평가서 5개팀 중 최하점 nc AI

주요 일정 및 향후 계획

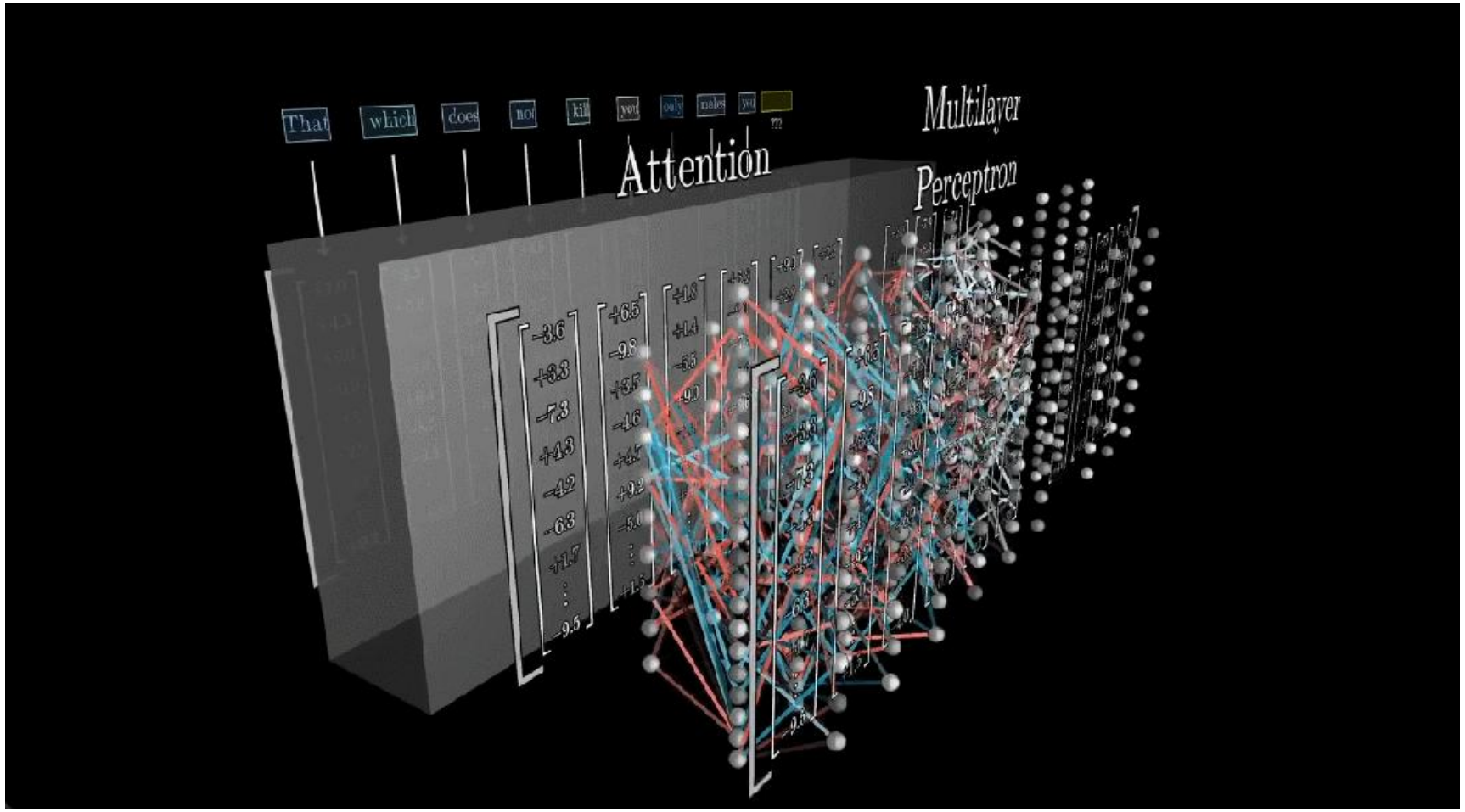
- **2025년 8월**
  5개 정예팀 선발, 프로젝트 착수
- **2025년 12월**
  - 5개 정예팀 제출 모델, 에포크AI ‘주목할 만한 AI’ 등재
- **2026년 1월**
  - 정예팀 3개 팀 압축
- **2026년 상반기 중**
  - 탈락 팀, 신규 역량 기업 대상으로 1개 팀 추가 공모
  - 4개 팀 경쟁 체제 구축 예정
- **2026년 말**
  - 정예팀 2개 팀 최종 선발

자료: 과학기술정보통신부

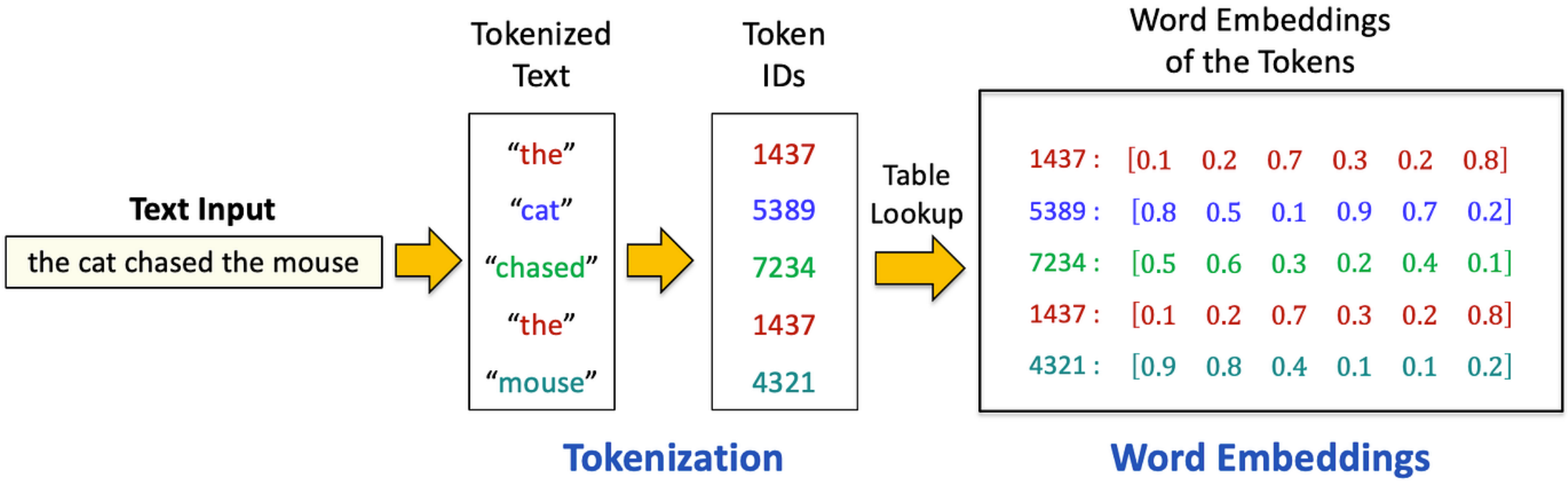
뉴시스

26.01.15 전진우 기자 618lue@newsis.com

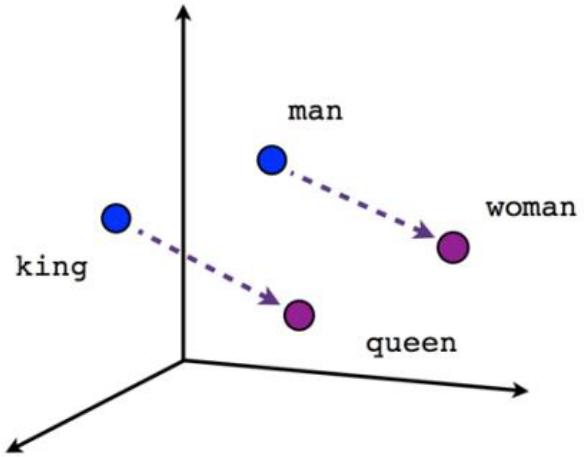
LLM Training = Training Transformer Architectures



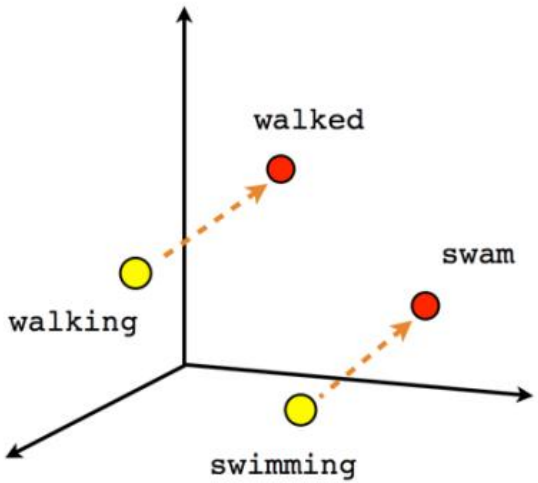
Tokenization and Word Vocab



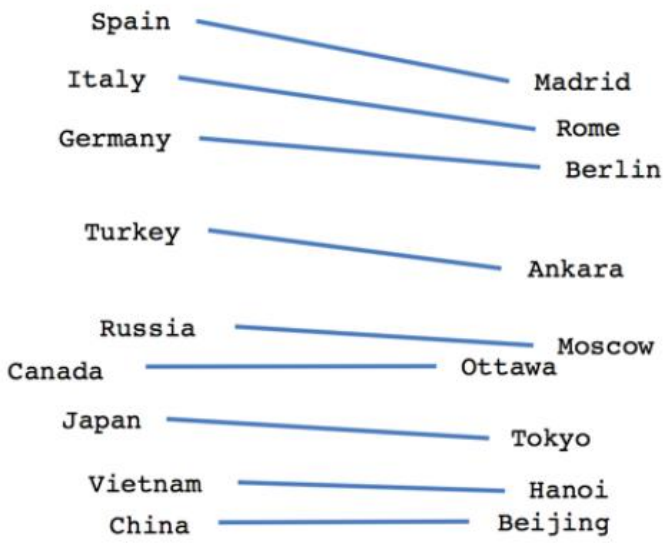
Tokenization and Word Vocab



Male-Female

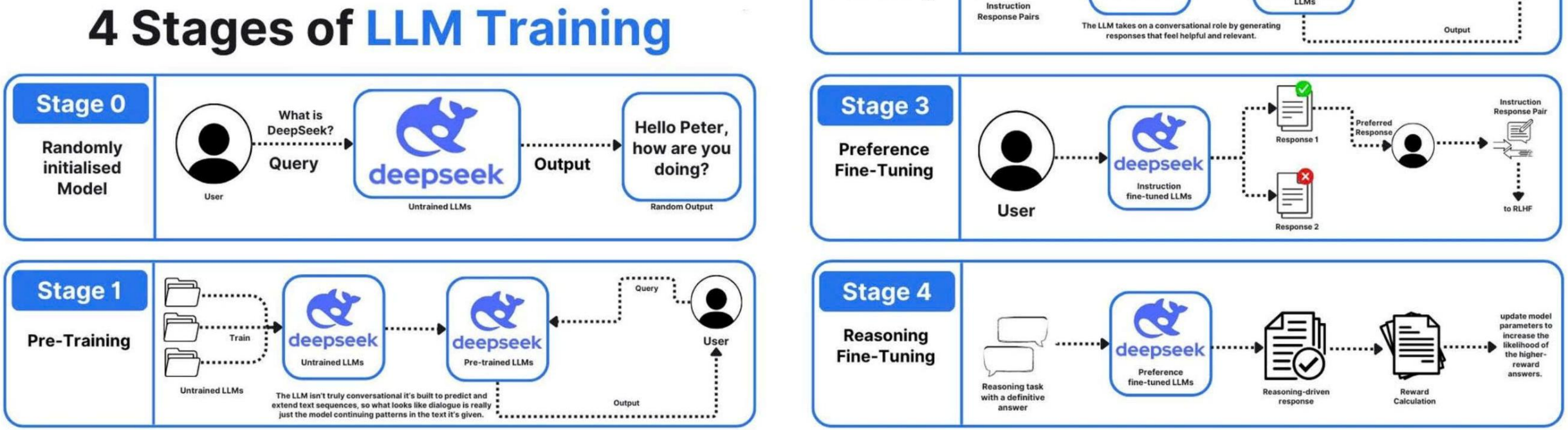


Verb tense



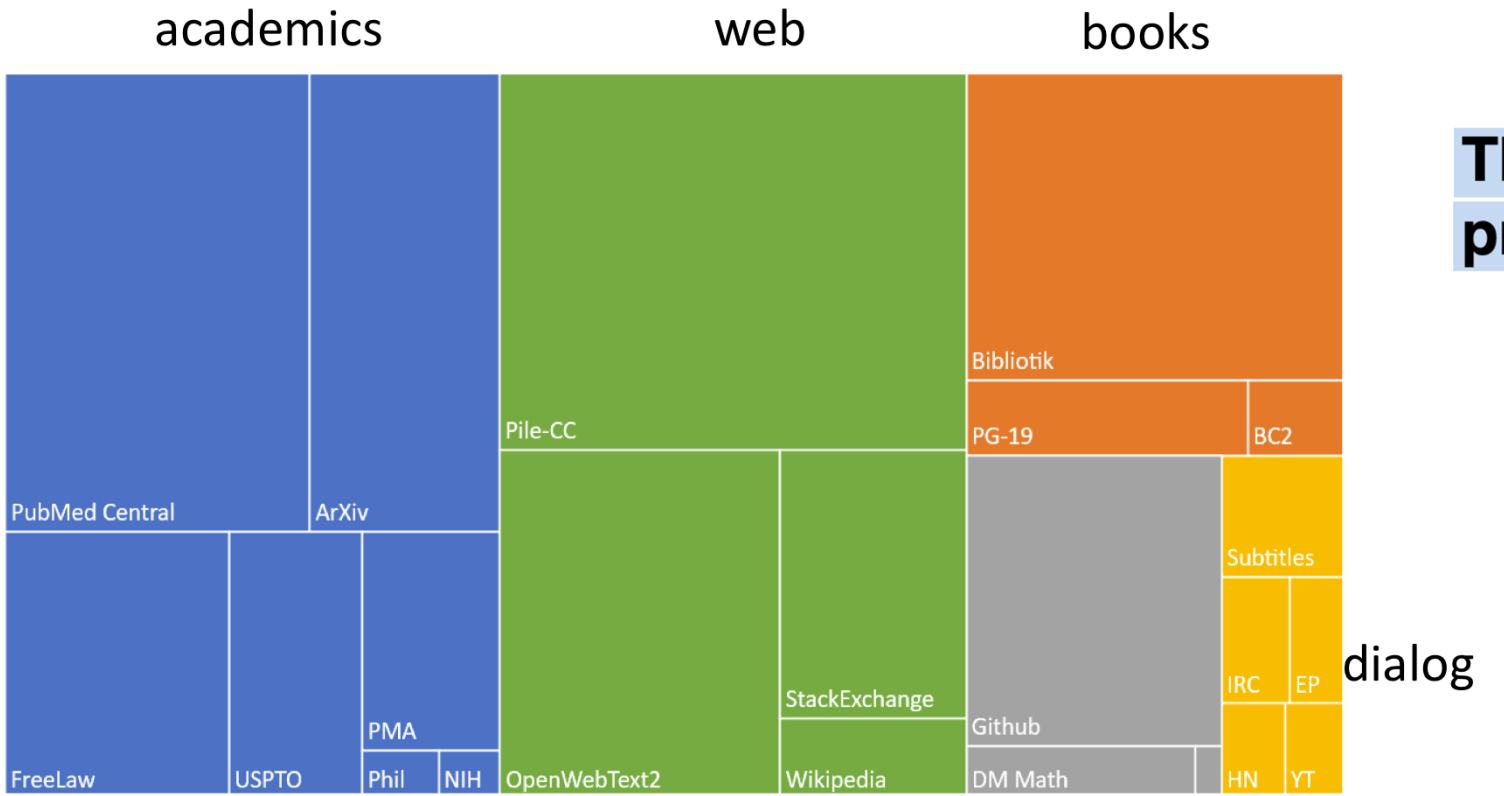
Country-Capital

Stages of LLM Training (3 or 4)



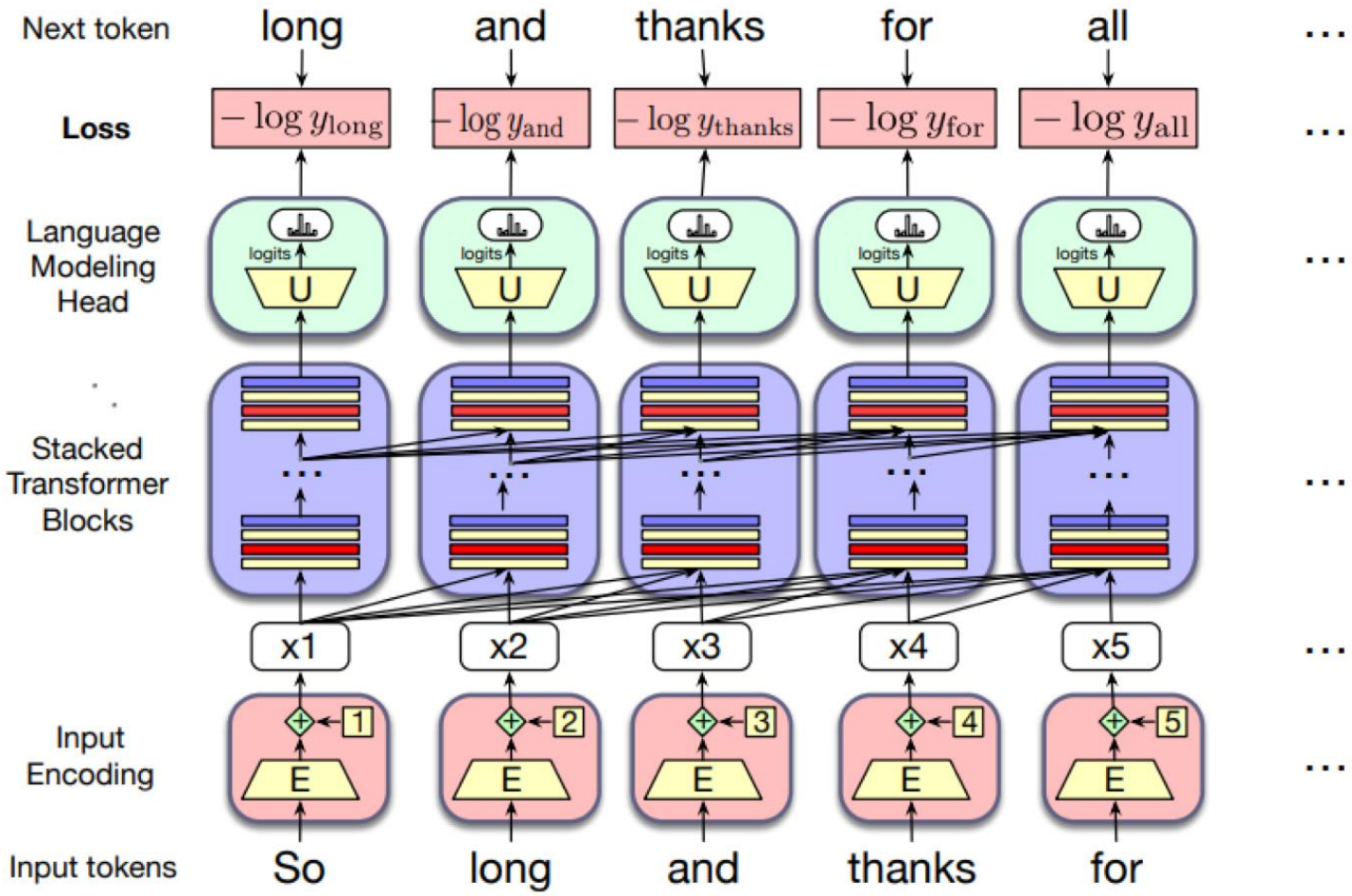
Pre-Training (Next Word Prediction) → **Fine-Tuning** (e.g., Instruction) → **Post-Training** (e.g., Preference, Reasoning)

Step1. Pre-Training



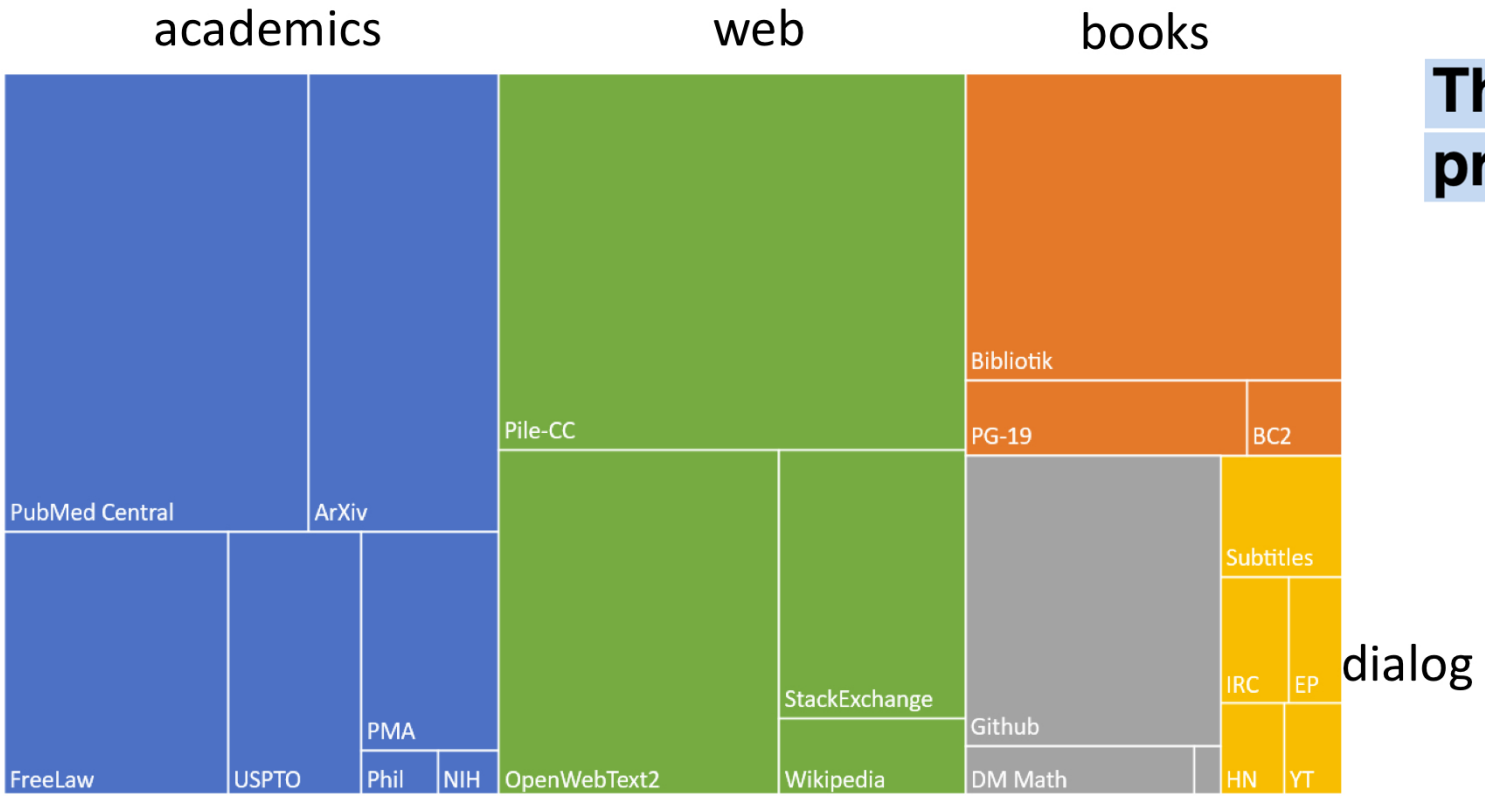
The model is simply trained to predict the next word

Step1. Pre-Training



$$= \frac{1}{T}\sum_{t=1}^{T} L_{CE}$$

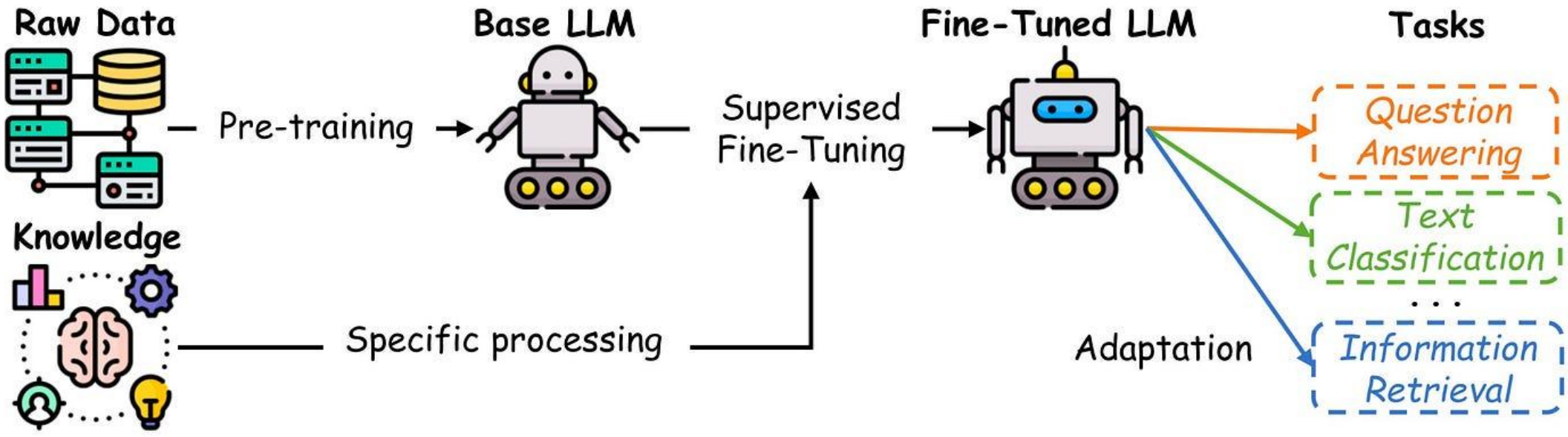
Step2. Fine-Tuning, Continued Pre-Training?



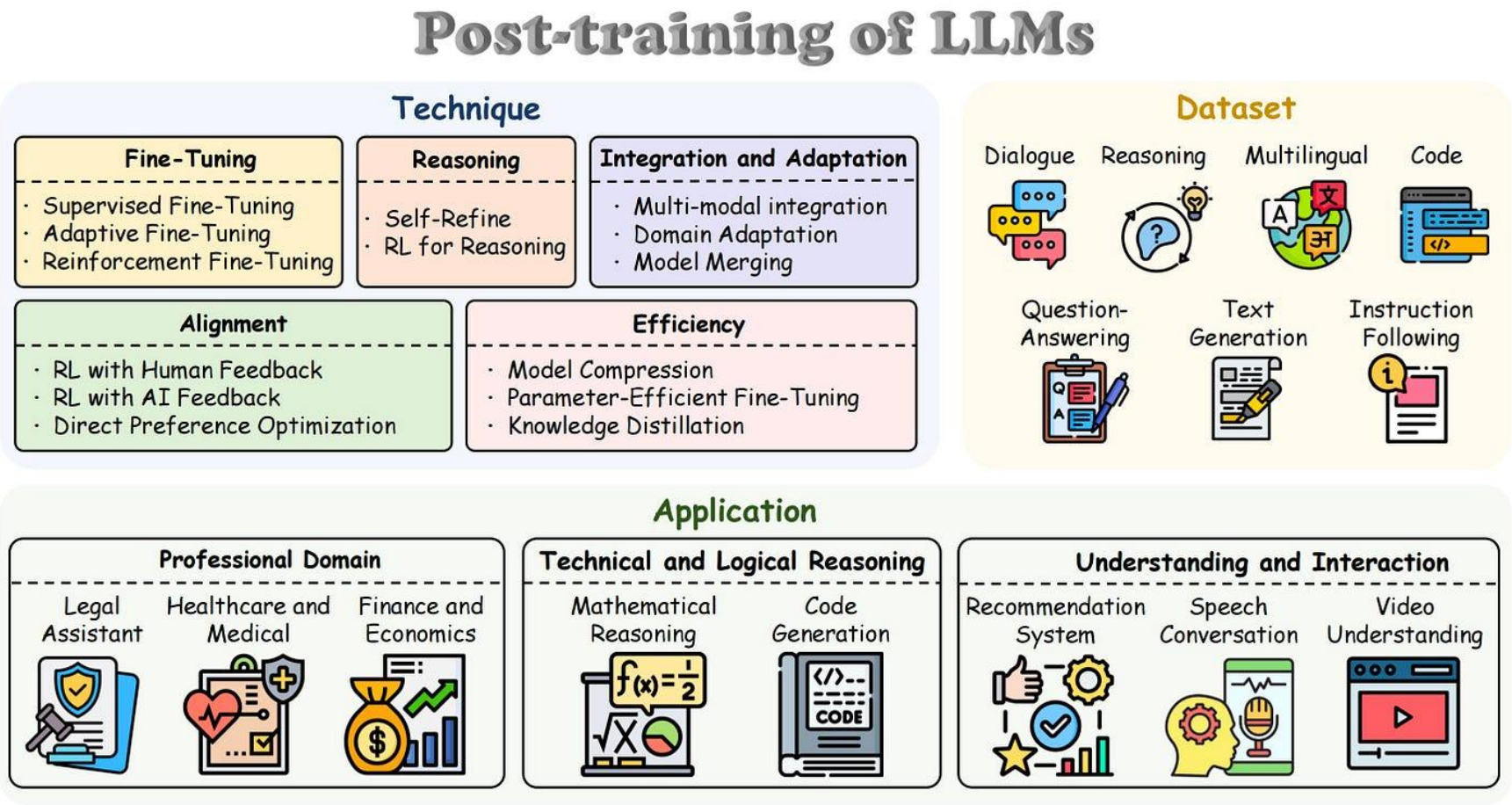
The model is simply trained to predict the next word

e.g., +Economics?

Step2. Fine-Tuning, Supervised Fine-Tuning (SFT) & Instruction-Tuning



Step3. Post-Training, Preference Alignment & Reasoning Capability



3. Understanding LLM Paradigm Again

Again, LLM (Training) Paradigm

LLM Training Objective	Keywords
Next Word Prediction (2017~)	Scaling Laws In-Context Learning (Few-Shot) Chain-of-Thought Prompting
Instruction Following (2021~) Preference Alignment (2022~)	Instruction-Tuning Supervised Fine-Tuning RLHF, DPO
Reasoning Capability (2024~)	GRPO, RLVR Inference/Test-Time Scaling Reasoning

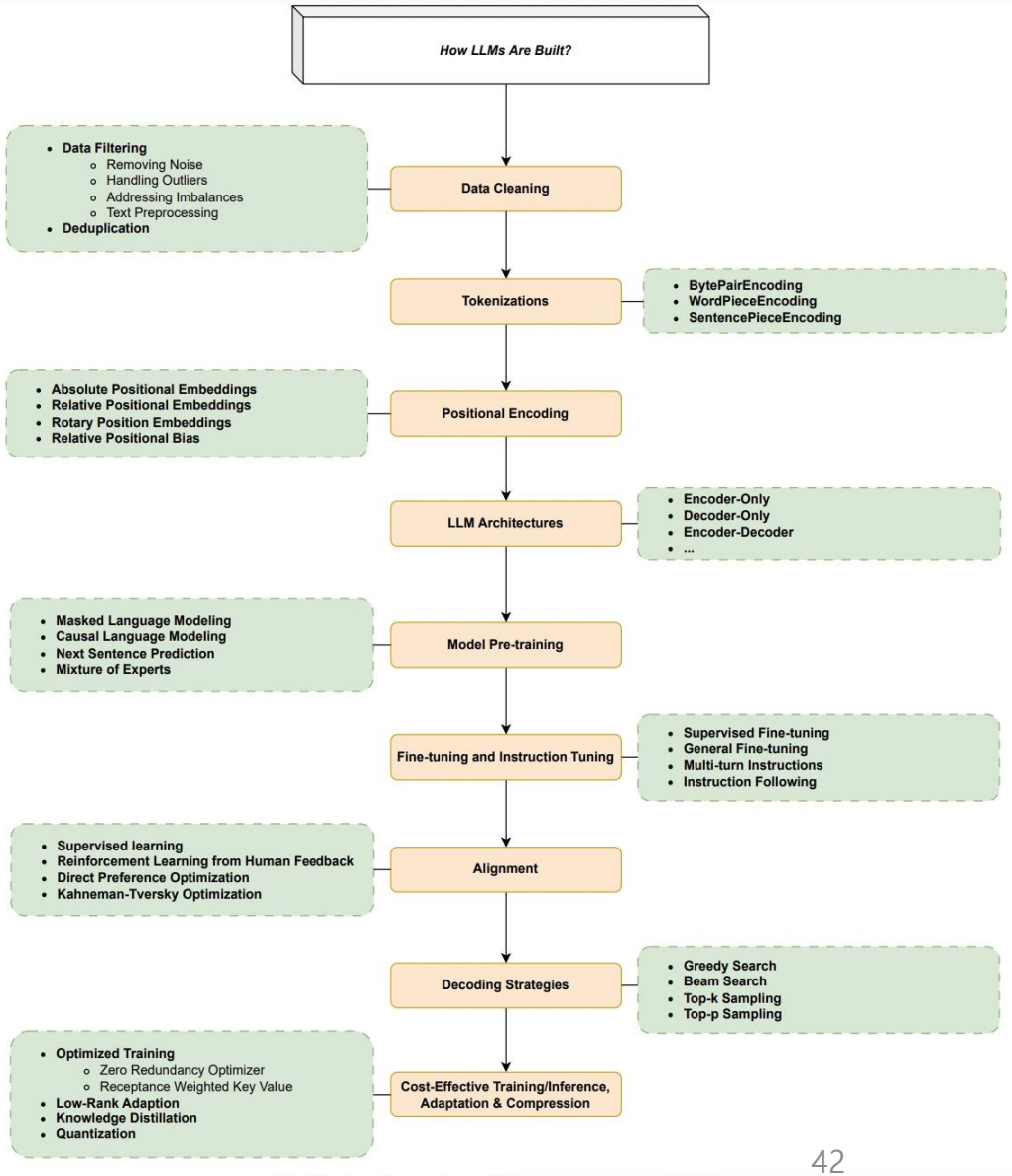


Fig. 25: This figure shows different components of LLMs.

4. More about LLM Reasoning

Various Themes for LLM Reasoning

Theme1. Reinforcement Algorithms for LLM Reasoning

Theme2. How to Improve LLM Chain-of-Thought Reasoning

Theme3. LLM Reasoning Domain (Math, Coding, Multimodal, …)

Theme4. Limitations and Potentials of LLM Reasoning

5. Now and Discussions

SFT vs RL

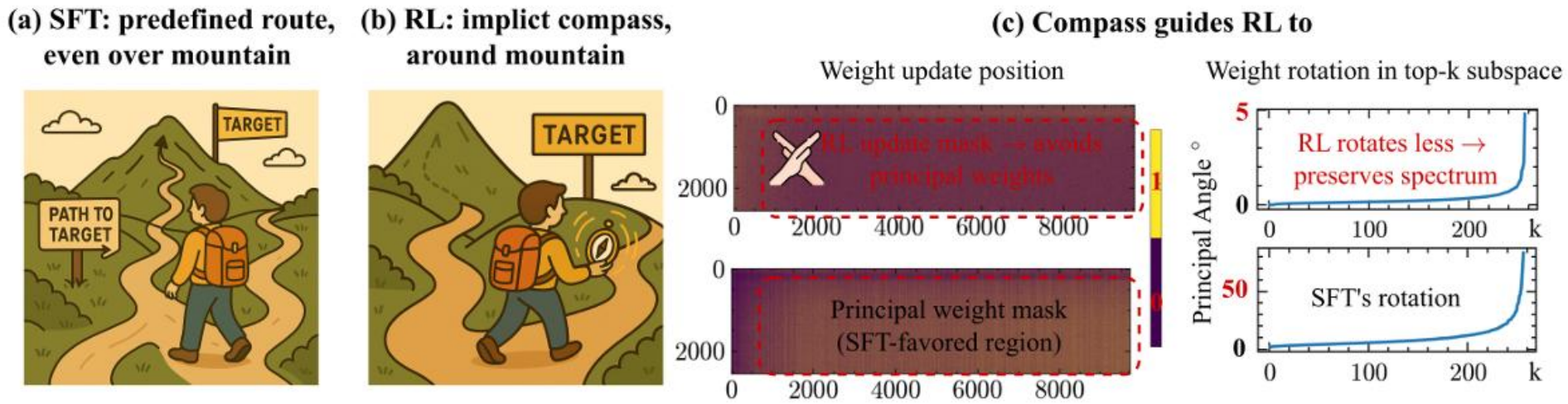
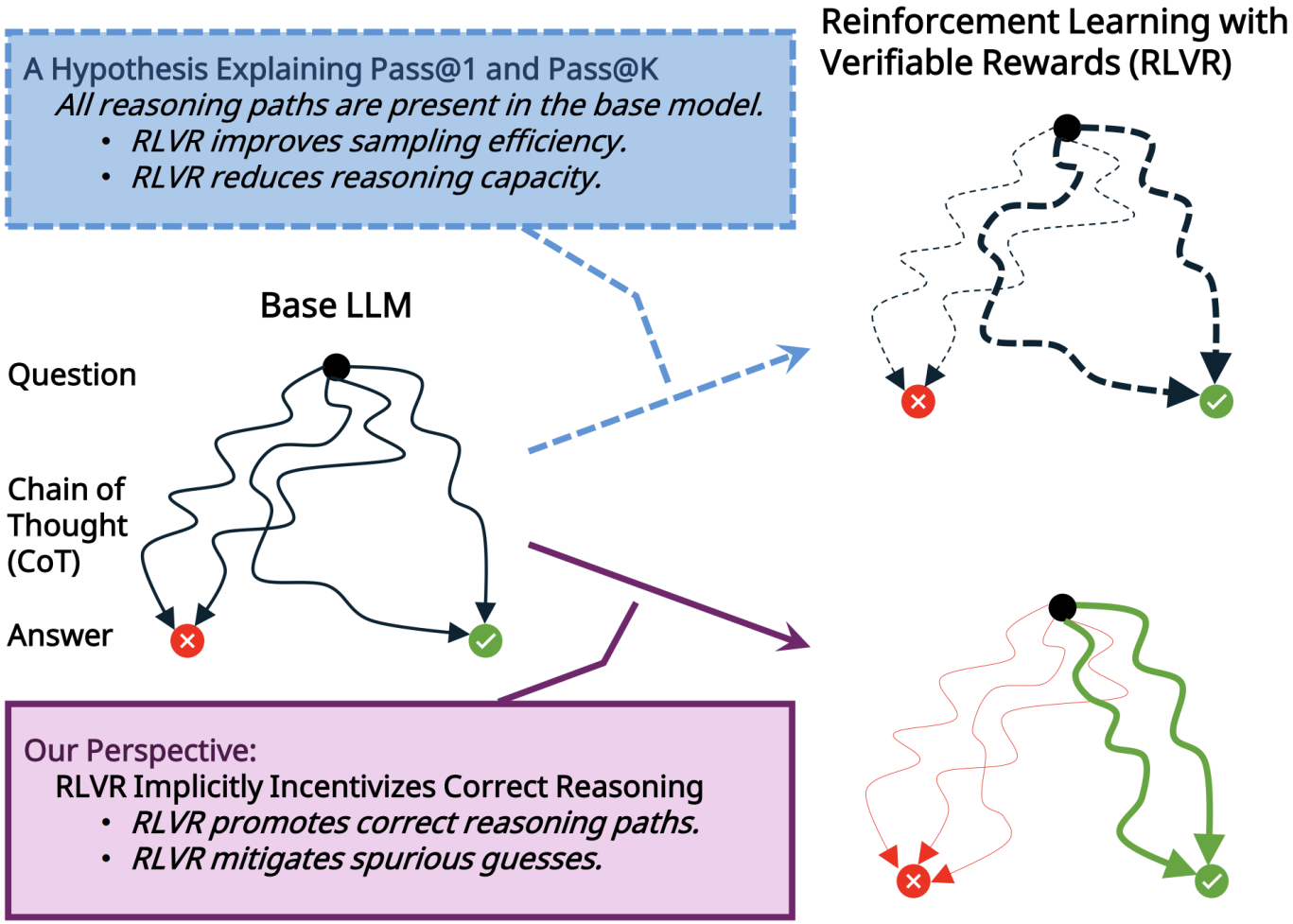


Figure 1 SFT vs. RLVR: optimization geometry and evidence. (a) **SFT** follows an externally guided route and traverses high-curvature directions ("over the mountain") to reach the target. (b) **RLVR**, without an explicit teacher, behaves as if steered by an *implicit compass* (a model-conditioned optimization bias), taking a low-curvature detour. (c) **Evidence.** *Left:* positional maps comparing the *update mask* (non-zero parameter updates) with the *principal mask* (positions aligned with top-k singular subspaces, defined by the largest-magnitude entries of the rank-k SVD reconstruction Liu et al. (2025c); details in Sec. 4.2). RLVR updates avoid principal-weight positions, whereas SFT targets them (Meng et al., 2024a; Liu et al., 2025c). *Right:* principal-angle curves of the top-k subspaces show that RLVR rotates less (spectrum preserved), while SFT rotates more.

Reasoning Capacity Boundary



Q&A